

Intensive Care Unit Management of Patients on Mechanical Circulatory Support

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KEY POINTS

1. Intra-aortic balloon pump counterpulsation increases coronary artery perfusion pressure, increases coronary artery blood flow (particularly in cardiogenic shock), and augments cardiac output by approximately 5% to 15%.
2. Catheter-mounted ventricular assist devices (VADs) are available for isolated right ventricular and left ventricular support. Left ventricular support devices can provide between 2.5 and 5.5 L/min of blood flow, depending on the product. Echocardiography should be performed daily when possible to confirm correct VAD position.
3. For left-sided catheter-mounted VADs (e.g., Impella 2.5, 5.0, CP) the inflow should be positioned approximately 3.5 cm below the aortic valve. For the Impella 5.5, the inflow should be positioned approximately 5 cm below the aortic valve.
4. Starting settings for the total artificial heart are left ventricular drive pressure of 180 to 200 mm Hg, right ventricular drive pressure of 30 to 60 mm Hg, heart rate of 100 to 120, and 50% systole. No native cardiac rhythm is required for proper device function.
5. The incidence of true heparin-induced thrombocytopenia (HIT) appears to be low during mechanical circulatory support. The 4T score can be used to categorize patients as low, intermediate, and high probability as in nonmechanical circulatory support patients.
6. Health care-associated infections are common in mechanical circulatory support patients, but there is no strong evidence to support use of prophylactic antibiotics. When signs of infection are present, targeted cultures should be performed, empiric antibiotics should be initiated, and tailored based on culture results.
7. A hemoglobin transfusion trigger of 7 g/dL is appropriate for most mechanical circulatory support patients, but exceptions may be appropriate in some clinical scenarios. There is no strong evidence to guide appropriate platelet transfusion during mechanical circulatory support. In extracorporeal membrane oxygenation patients with bleeding, it is reasonable to maintain a platelet count at least $>50 \times 10^9/L$ and perhaps $>100 \times 10^9/L$ because of qualitative platelet dysfunction.
8. Early mobility should be promoted as soon as is feasible in mechanical circulatory support patients, even those with peripheral cannulation. Mobility helps maintain strength, reduces muscle atrophy, and improves patient experience. A multidisciplinary approach is necessary to limit complications during ambulation and physical therapy.
9. Complex ethical dilemmas can arise in temporary mechanical circulatory support patients. Early discussions with patients and families about what temporary mechanical circulatory support can and cannot accomplish are critical to help create reasonable expectations.

Introduction and Historical Context

In the early 2000s, a renaissance in mechanical circulatory support (MCS) occurred with multiple new temporary and durable devices entering clinical practice. By 2018, cardiogenic shock mortality fell to a historic low of 35%.¹ Today, multiple MCS devices are available for right ventricular (RV) support, left ventricular (LV) support, or biventricular support. At present, the only durable left ventricular assist device (LVAD) that is commercially available in the United States is the HeartMate 3 (Abbott Cardiovascular, Plymouth, MN). In early 2021, the HVAD (HeartWare, Framingham, MA), which had been implanted in more than 19,000 patients worldwide was removed from the market by its manufacturer because of concerns about its high stroke rate and other factors.²

Indications for Mechanical Circulatory Support

Patients with refractory cardiogenic shock should be evaluated for MCS unless they have other end-stage diseases (e.g., metastatic cancer) or are of

extreme age. There should be consideration of whether temporary MCS will be used as a bridge to recovery, transplantation, or durable LVAD implantation. There is no universal definition for cardiogenic shock, but accepted criteria include impaired LV or RV systolic function with corroborating findings such as (1) sustained systolic blood pressure <90 mm Hg or mean arterial pressure (MAP) <65 mm Hg; (2) pulmonary congestion with elevated LV end-diastolic pressure (>16 mm Hg); (3) signs of organ malperfusion; (4) cardiac index <2.2 L/min/m²; (5) mixed venous oxygen saturation (SvO₂) less than 60%; or (6) elevated biomarkers indicative of tissue hypoxemia (e.g., elevated blood lactate concentration).^{3,4} Low cardiac power output (cardiac output [CO, L/min] $\times$ MAP [mm Hg] $\times 0.0022$) is indicative of cardiogenic shock. Patients with cardiac power output <0.5 have mortality that exceeds 50% and should be considered for MCS if they do not improve with inotropes.^{5,6}

Cardiogenic shock is initially treated with inotropes with a goal to increase MAP >65 mm Hg and improve RV and/or LV systolic function (see [Chapter 34](#)). Other interventions include metabolic optimization (e.g., optimization of mechanical ventilation and treatment of hypermetabolism) and optimization of tissue perfusion (e.g., reduction in central venous pressure [CVP]) ([Fig. 35.1](#)). When cardiogenic shock

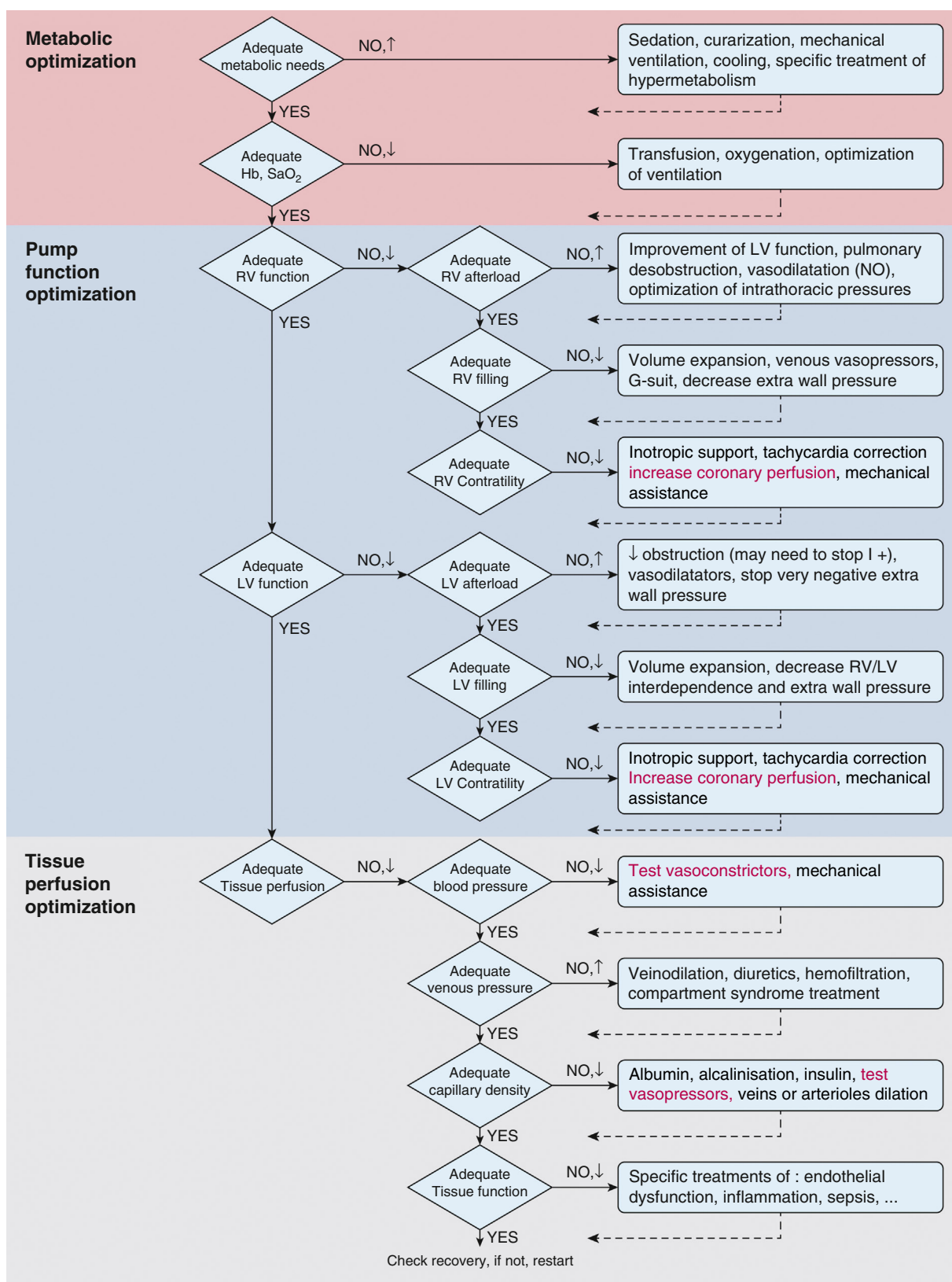


Figure 35.1 Sequential management steps for managing cardiogenic shock. Hb, Hemoglobin; LV, left ventricle; NO, nitric oxide; RV, right ventricle; SaO₂, arterial oxygen saturation. (From Squara P, Hollenberg S, Payen D. Reconsidering vasopressors for cardiogenic shock: everything should be made as simple as possible, but not simpler. Chest. 2019;156(2):392-401.)

cannot be reversed with inotropes and medical interventions, MCS should be considered. Early MCS is associated with superior outcomes when compared to using high-dose vasopressors and/or inotropes (e.g., norepinephrine $> 0.2 \mu\text{g/kg/min}$).⁷ The initial modality depends on multiple factors, but is mostly related to whether isolated RV, LV, or biventricular support is needed, what vascular access sites are available, and the shock team's experience with various devices. Although the intra-aortic balloon pump (IABP) was historically the "gold standard" for initial MCS, there has been a steady trend toward using devices that can provide higher CO augmentation. In a 2015 expert consensus statement, catheter-mounted axial continuous-flow VADs and venoarterial (VA) extracorporeal membrane oxygenation (ECMO) were recommended as first-line MCS therapies, except in patients with myocardial infarction (MI) who could quickly be revascularized, in whom an IABP might be equivalent.³

Mechanical Circulatory Support Devices

Intra-aortic Balloon Pump

IABP counterpulsation was introduced by Kantrowitz in the 1960s.⁸ Using a percutaneous Seldinger technique, an IABP (usually 7–8 Fr catheter) is placed through a large artery, usually the common femoral artery, into the descending thoracic aorta with or without a vascular introducer sheath. IABP placement can be performed at the bedside, in the operating room, or in the cardiac catheterization laboratory where fluoroscopy can be used to confirm correct position. Chest x-ray and transesophageal echocardiography (TEE) can be used to confirm correct IABP position when fluoroscopy is not available. Modern IABP catheters have a polyurethane cylindrical balloon that inflates with helium during diastole and deflates during systole. They also have a fiberoptic tip that allows for rapid transfer of arterial pressure data to the console and in vivo autocalibration.

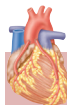
IABP counterpulsation has two main hemodynamic benefits (Box 35.1). First, coronary artery perfusion pressure increases during diastole when the IABP inflates. Because coronary blood flow is autoregulated, increased perfusion pressure does not guarantee increased coronary blood flow. Increased coronary artery perfusion pressure only increases coronary blood flow when autoregulation is impaired, as in shock.⁹ Second, cardiac afterload is reduced during systole when the IABP deflates. Rapid IABP deflation increases stroke volume and reduces myocardial stroke work.¹⁰ In animal studies, this modestly increases CO (5%–10%) in the absence of shock, while there is an increase of up to 15% in shock (Fig. 35.2).¹¹ The alternating pattern of IABP inflation and deflation that

occurs leads to a characteristic arterial waveform (Fig. 35.3), which has two pressure peaks or a "bunny ear" pattern.

The strongest evidence to support IABP use is in patients having complex percutaneous coronary interventions (PCIs) and in patients with large ST elevation MIs (STEMIs) (see Chapter 2). The balloon pump–assisted coronary intervention study, which included over 300 patients having complex PCI, demonstrated a long-term survival advantage in patients who had an IABP placed before PCI (Fig. 35.4).¹² The Counterpulsation to Reduce Infarct Size pre-PCI acute MI study (CRISP-AMI) found that STEMI patients who were randomized to have an IABP placed had significantly less death, shock, and new congestive heart failure (CHF) (5% vs 12% when compared to controls).¹³ In this study, the benefit of IABP counterpulsation was particularly high in patients with large MIs (ST elevation $\geq 15 \text{ mm}$).¹⁴ Of note, in the landmark IABP in cardiogenic shock II trial (IABP-SHOCK II), which included 600 patients, both non-ST elevation MI (NSTEMI) and STEMI patients with cardiogenic shock were included and there was no reduction in short or long-term mortality with IABP placement at the time of PCI.^{15,16} Recently, there has been a trend toward using more axillary IABPs, as this configuration allows patients with decompensated CHF or cardiogenic shock to stand, ambulate, and improve their priority on the cardiac transplant allocation list to Status 2.

Catheter-Mounted Axial Continuous-Flow Ventricular Assist Devices

Catheter-mounted axial continuous-flow VADs are designed to have a miniaturized blood pump as part of the catheter. Because of their design, the inflow and outflow must be inline, and hence axial-flow is



BOX 35.1 KEY ASPECTS OF INTRA-AORTIC BALLOON PUMP (IABP)

- Inflates during diastole and deflates during systole
- Three pressures measured: assisted systolic, assisted end-diastolic, and balloon-augmented diastolic
- Late deflation and early inflation of IABP increase myocardial work
- Late inflation and early deflation of IABP reduce its efficacy
- Highest level of evidence for IABP is in complex percutaneous coronary intervention and in patients with ST elevation myocardial infarction
- IABP increases coronary artery perfusion pressure and coronary artery blood flow, especially in cardiogenic shock

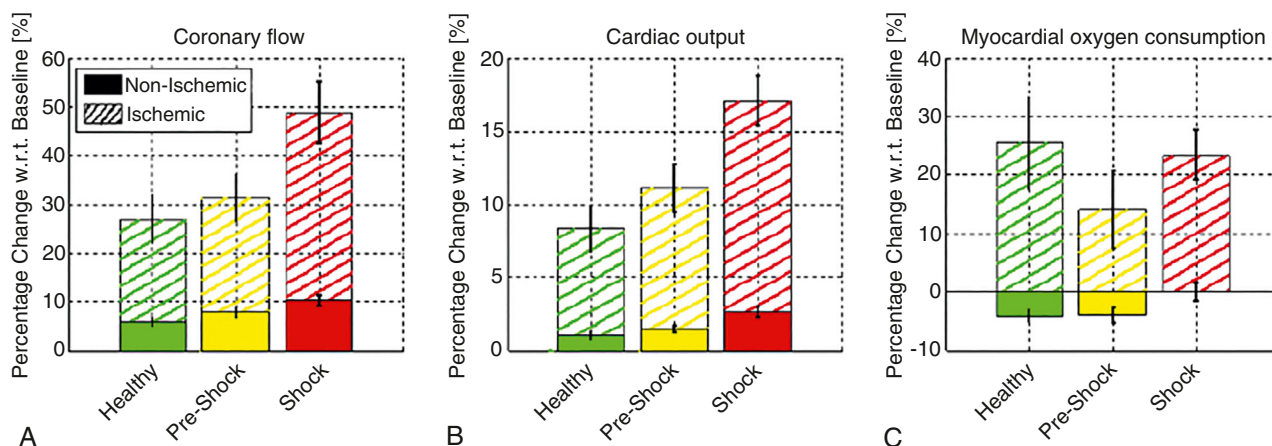


Figure 35.2 Hemodynamic effects of intra-aortic balloon pump counterpulsation in pigs for (A) coronary flow, (B) cardiac output, and (C) myocardial oxygen consumption. (From van Nunen LX, Noc M, Kapur NK, Patel MR, Perera D, Pijls NH. Usefulness of intra-aortic balloon pump counterpulsation. *Am J Cardiol*. 2016;117:469-476.)

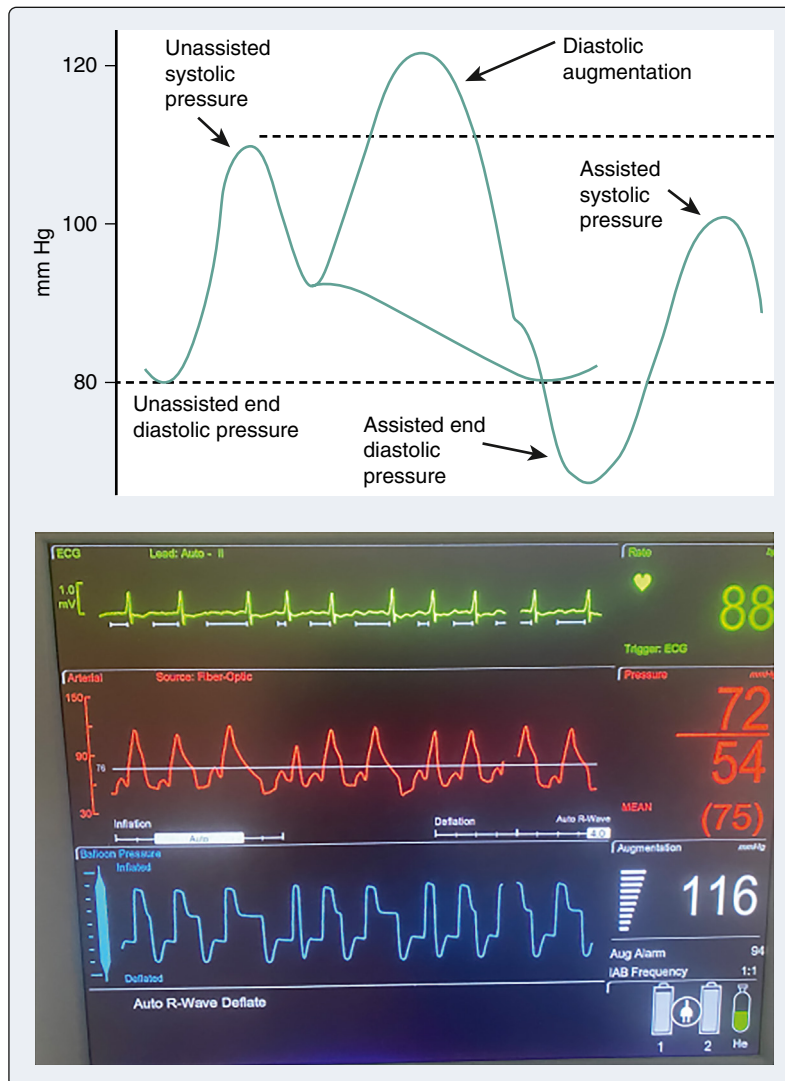


Figure 35.3 (Top) The waveform shows pressures that are displayed on the intra-aortic balloon pump console; diastolic augmented, assisted diastolic, and assisted systolic. (Bottom) The display on the intra-aortic balloon pump console shows the patient's electrocardiogram, arterial waveform, and a pressure tracing for balloon. Red numbers show assisted systolic and diastolic pressures. The white number shows diastolic augmented pressure.

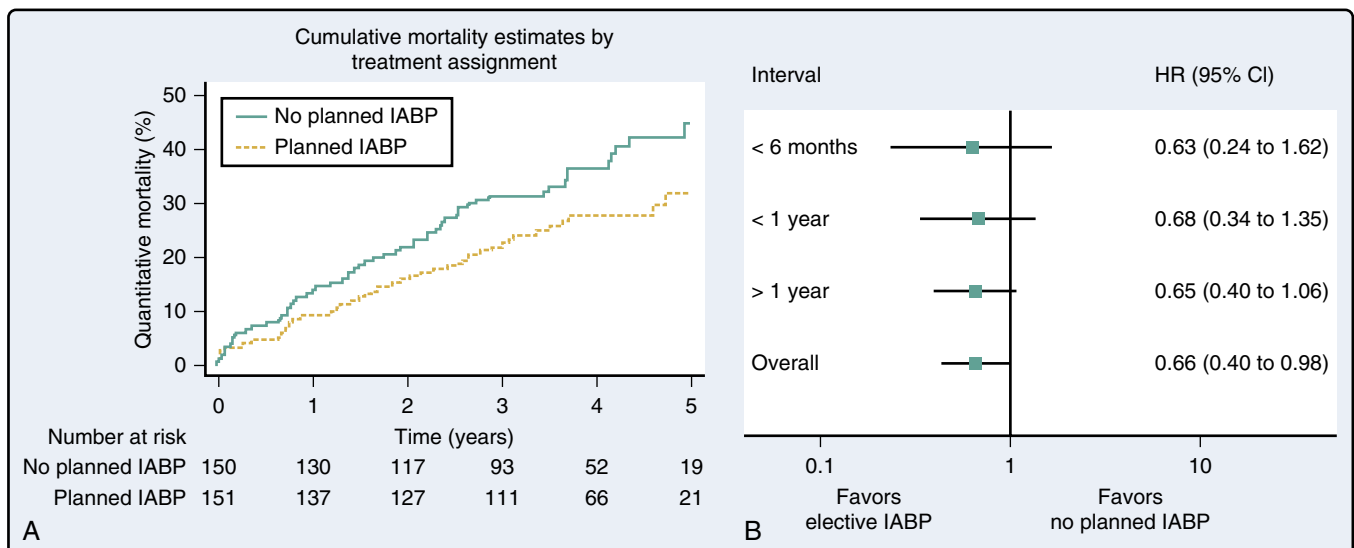


Figure 35.4 Impact of an intra-aortic balloon pump (IABP) on survival in patients having complex percutaneous coronary interventions. Panel A shows cumulative mortality over time in patients with and without a planned IABP. Panel B shows hazard ratios for mortality at various timepoints. CI, Confidence interval; HR, hazard ratio. (Redrawn from van Nunen LX, Noc M, Kapur NK, Patel MR, Perera D, Pijls NH. Usefulness of intra-aortic balloon pump counterpulsation. *Am J Cardiol*. 2016;117:469-476.)

mandated. In contrast, radial-flow pumps have an orthogonally oriented inflow and outflow. The Impella (Abiomed, Danvers, MA) line of catheter-mounted axial continuous-flow VADs is the only Food and Drug Administration (FDA)-approved system in the United States (Fig. 35.5). Various pump sizes have proportional blood flow capacities for a given pump speed and afterload. It is important to note that axial-flow VADs have a linear inverse relationship between the transpump pressure difference, which when divided by the product of fluid density and acceleration of gravity yields the “head pressure,” and VAD flow. Head pressure is the primary determinant of VAD flow at any point in the cardiac cycle. In contrast, radial-flow pumps, often referred to as “centrifugal-flow pumps,” have a nonlinear head pressure–blood flow relationship.

VADs capable of maximum blood flows of 2.5, 3.5 (CP), 5.0, and 5.5 L/min are available (Fig. 35.5 and Table 35.1), with the Impella RP, available for RV support (Box 35.2). The smaller devices (2.5, CP, and RP) are most commonly placed percutaneously, via transfemoral approaches. In contrast, the “full-support” 5.0 and 5.5 devices require surgical placement. Generally, this is via a side graft technique. With such an approach, surgical exposure of a systemic artery is performed. A side artery to end Dacron graft anastomosis is then created using a polypropylene suture. A temporary introducer sheath for the device is placed through the free end of the graft. The guidewire and catheter are advanced retrograde, crossing the aortic valve, and into the LV cavity. The catheter is removed, and the VAD is advanced into correct position over the guidewire. The guidewire is removed and device support is initiated. Placement of full-support devices can be performed peripherally (e.g., axillary artery) or centrally via the ascending thoracic aorta.

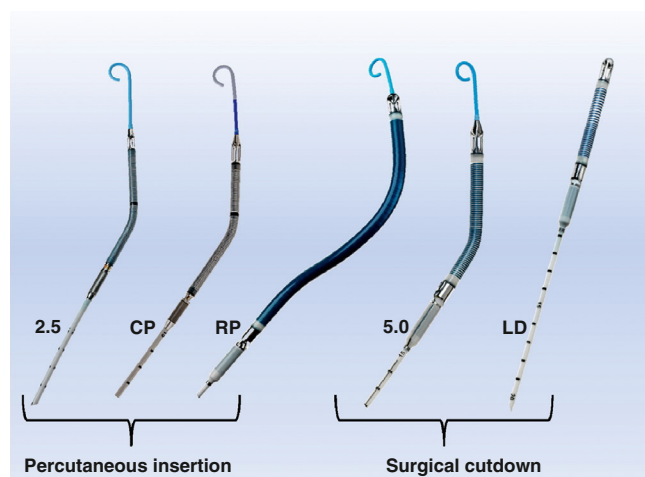


Figure 35.5 Various Impella ventricular assist devices. (From Burzotta F, Trania C, Doshi SN, et al. Impella ventricular support in clinical practice: collaborative viewpoint from a European expert user group. *Int J Cardiol.* 2015;201:684-691.)

Impellas have 10 speed settings (P0–P9), compared to the greater number of pump speeds and smaller stepwise titrations available with large platform devices. The blood flow and extent of LV unloading is proportional to the P number. For example, for the 5.5 device, P0 is 0 revolutions per minute (RPM), P1 is 12,000 RPM, P2 is 17,000 RPM, and each further iteration increases RPM by 2000 to 3000.

Impellas have been used extensively to support patients with cardiogenic shock in the post-MI setting, in postcardiotomy shock, and as a venting strategy in VA ECMO patients. In a systematic review that included over 600 cardiogenic shock patients from 11 studies, complication rates were relatively low with a major bleeding rate of 19.9%, hemolysis rate of 10.5%, limb ischemia rate of 5.0%, and stroke rate of 3.8%.¹⁷ When evaluating only high-risk PCI patients (over 7000 patients), complication rates were lower with a major bleeding rate of 6%, stroke rate of 2.8%, and vascular complication rate of 2.5%.¹⁸ When used as an adjunct for LV venting in VA ECMO patients, Impellas may reduce mortality by up to 50%, but prospective randomized trials are needed to confirm these findings.¹⁹

Paracorporeal Continuous-Flow Ventricular Assist Devices With Percutaneous or Peripheral Cannulation

While catheter-mounted VADs are capable of generating relatively high blood flows, miniaturization leads to high shear stress, which increases blood trauma. Larger pumps can achieve equivalent blood flow with lower pump speeds and higher maximum blood flow. In cases where patients require a high level of circulatory support (e.g., blood flow > 4–5 L/min) these features may be advantageous.

Peripheral cannulation can be used with paracorporeal VADs. For LV support, the archetypal example is the TandemHeart LVAD (TandemLife, Pittsburgh, PA). To place the device's inflow, a long femoral venous cannula is placed through a common femoral vein, up the inferior vena cava (IVC) and across the atrial septum into the left atrium (LA). The device's outflow is placed into a common femoral artery. The proprietary blood pump is frequently used, but other blood pumps can be employed in concert with the cannulas. The TandemHeart LVAD has been used both to support patients in cardiogenic shock and patients having high-risk PCI procedures.²⁰ The device is easily inserted and can provide blood flow up to 4 L/min at a maximum RPM of 7500.

For isolated RV support, a single cannula approach can be employed. A dual-lumen right ventricular assist device (RVAD)-ECMO cannula (Fig. 35.6), draws blood from the right atrium (RA) through three sets of inflow holes and returns it into the main pulmonary artery (PA). These devices have been used both to manage postcardiotomy RV failure and non-surgical RV failure from multiple causes including acute respiratory distress syndrome (ARDS), pulmonary embolism (PE), and acute MI.^{21–23}

Venoarterial Extracorporeal Membrane Oxygenation

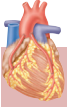
VA ECMO can be used to support patients with cardiogenic shock from multiple etiologies including end-stage heart failure, acute MI, PE,

TABLE 35.1 Features of the Various Impella Ventricular Assist Devices

	Impella 2.5	Impella CP	Impella 5.0/LD	Impella RP
Access	Percutaneous, femoral	Percutaneous, femoral	Surgical, axillary/fem or ascend aorta	Percutaneous, femoral vein
Output (max; L/min)	2.5	4.0	5.0	4.6
Guiding catheter size (Fr)	9	9	9	11
Motor size (Fr)	12	14	21	22
Introducer size	13 Fr peel away	14 Fr peel away	Dacron graft 10 mm recommended	23 Fr peel away
RPM (max)	51,000	46,000	33,000	33,000
EU approval	5 days CE mark	5 days CE mark	10 days CE mark	14 days CE mark

EU, European Union; RPM, revolutions per minute.

From Burzotta F, Trania C, Doshi SN, et al. Impella ventricular support in clinical practice: collaborative viewpoint from a European expert user group. *Int J Cardiol.* 2015;201:684-691.



BOX 35.2 KEY ASPECTS OF CATHETER-MOUNTED AXIAL CONTINUOUS-FLOW VENTRICULAR ASSIST DEVICES (VADs)

- Devices available for right ventricular and left ventricular support
- Left-sided devices include Impella 2.5, 5.0, 5.5, CP, and LD
- Right-sided device is the Impella RP
- Daily echocardiography should be performed when feasible to confirm correct device position
- The inflow should be approximately 3.5 cm below the aortic valve for the Impella 2.5, 5.0, CP, and LD
- The inflow should be approximately 5 cm below the aortic valve for the Impella 5.5
- Placement signal varies by device, with Impella 5.0, LD, and RD having a differential pressure sensor, Impella 2.5 and CP having a fluid filled pressure sensor, and Impella 5.5 having a fiberoptic sensor
- Impellas have 10 speeds between P0 and P9 and blood flow increases proportionally with pump speed
- Blood flow is inversely related to the head pressure (inflow cannula pressure – outflow cannula pressure), with blood flow being highest during systole and lowest during diastole



Figure 35.6 The ProtekDuo cannula. Inflow orifices are near the operator's fingers and outflow orifices are near the end of the cannula.

drug intoxication, and refractory ventricular arrhythmia. It provides biventricular support, reduces cardiac preload, and increases afterload modestly. VA ECMO can be performed centrally (e.g., RA-ascending aorta) or peripherally (e.g., common femoral vein-common femoral artery). VA ECMO management is covered extensively in [Chapter 27](#).

Total Artificial Heart

The CardioWest total artificial heart (Syncardia, Tucson, AZ) is a pneumatically driven biventricular VAD that has been implanted in more than 1800 patients worldwide ([Fig. 35.7](#)). It is approved by the United States FDA as a bridge-to-cardiac transplantation. The device has two polyurethane artificial ventricles that each have a 70-mL stroke volume and two

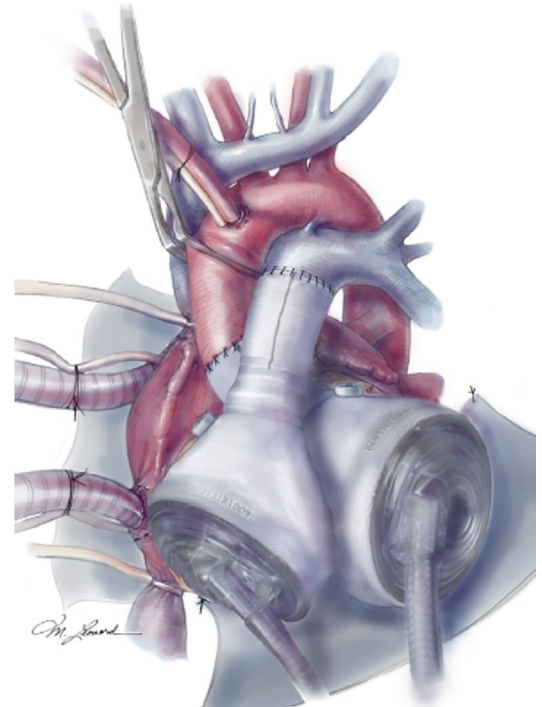


Figure 35.7 Total artificial heart device with two pneumatically driven ventricles. (From Goodwin ML, Mokadam NA. Total artificial heart implantation: how I teach it. *Ann Thorac Surg*. 2019;108:1271-1276.)



BOX 35.3 TOTAL ARTIFICIAL HEART CONSIDERATIONS

- Two pneumatic artificial ventricles
- Maximum stroke volume is 70 mL for each ventricle
- Inotropes not necessary, vasopressors used to maintain a normal mean arterial pressure
- No intrinsic cardiac rhythm necessary
- Can tolerate lower hemoglobin concentration of approximately 6 g/dL

tilting disk valves (one inflow and one outflow) ([Box 35.3](#)).²⁴ The 70-mL device can deliver a CO of up to 9.5 L/min.²⁵ Body size is an important consideration for patients who have total artificial heart implantation and a body surface area > 1.7 m² is required for the 70-mL device.²⁵ In addition, the anterior-posterior chest distance (from posterior sternum to vertebral body) should be at least 10 cm. The total artificial heart can be used to treat biventricular failure from multiple causes, and is particularly useful in patients with refractory ventricular arrhythmias, as no cardiac rhythm is required for the device to function properly.²⁵

Durable Left Ventricular Assist Devices

Durable LVAD management primarily with the HeartMate 3 is covered in detail in [Chapter 22](#).

Echocardiography for Temporary Devices

General Considerations

The American Society of Echocardiography maintains detailed recommendations for the use of echocardiography in MCS management.²⁶

Limitations include operator skill and experience and the quality of imaging windows.²⁷ The latter is impacted by body habitus, patient positioning, implanted hardware, chest tubes, or air in the subcutaneous, pleural or mediastinal spaces, which occurs commonly after cardiothoracic surgery.²⁸ In cases where transthoracic echocardiography (TTE) images are poor, TEE can produce superior images, but TEE is invasive and requires specialized training and certification.²⁹ Regardless of the modality that is used, it is useful to have baseline reference images for comparison, so that changes in loading conditions, RV and LV function, and device position can be assessed.

TTE or TEE can be used for pump speed adjustment and assessment of cardiac recovery during bedside weaning (ie, ramp study). Both RV and LV function can easily be assessed, as device speed is decreased and preload increases. With TTE, the apical four-chamber view is useful for assessing both RV and LV systolic function simultaneously. Systolic function should be assessed subjectively and using quantitative measures such as tricuspid annular plane systolic excursion (TAPSE), the myocardial performance index, tissue Doppler annular velocity, and fractional area change. Global longitudinal strain and three-dimensional assessment of LV ejection fraction can be performed with appropriate hardware and software packages. Quantitative measurements of RV and LV internal diameters at end-diastole should be performed to assess chamber dilation.

Tricuspid regurgitation (TR) and mitral regurgitation (MR) can be assessed from the TTE apical four-chamber view. If severe functional TR or MR occurs as pump speed is decreased, it can be a sign of poor RV or LV function. RV outflow tract (RVOT) and LV outflow tract (LVOT) velocity time integrals (VTI) can be used to trend changes in RV and LV stroke volume as pump speed is changed. The RVOT VTI can be measured from a TTE parasternal RVOT view or TEE transgastric RVOT view. The LVOT VTI can be measured from a TTE apical five-chamber view or TEE deep transgastric view.

Adequate ventricular decompression is critical in MCS patients to reduce metabolic demand and prevent pulmonary (LV failure) and abdominal organ congestion (RV failure). While invasive hemodynamic monitors (e.g., PA catheter) or the MCS device itself (e.g., Impellas) can measure ventricular pressures, echocardiography allows for visualization of chamber size, comparison of RV and LV diameter, and quantification of valvular regurgitation, making it an important adjunct to invasive hemodynamic monitoring.³⁰ When the LV is adequately decompressed, its end-diastolic diameter (LVEDD) should ideally be below the upper limit of normal (<5.5 cm). For a chronically dilated heart, this may not be possible, but current LVAD management guidelines recommend at least a 15% reduction in LVEDD with decompression. RV decompression can be assessed with TTE in the apical four-chamber or subxyphoid views. If RV or LV decompression is not adequate, potential corrective actions include increasing pump speed, LV venting during peripheral VA ECMO, or surgical closure of the aortic valve in patients with aortic insufficiency (AI) who have an LVAD.

Excessive unloading of the RV or LV can be problematic because it can cause ventricular suckdown. Suckdown can cause ventricular arrhythmias and also compromise ventricular function when the septal contribution to systole is impaired in the contralateral ventricle. In cases of suspected over-decompression or hypovolemia, TTE can be used to help adjust pump speed to a level where the interventricular septum is midline and there is no further suckdown.³¹ Ideally, the interventricular septum should be midline between the two ventricles when viewed in a TTE or TEE four-chamber view and in a normal position when the ventricles are viewed in a parasternal mid-papillary short-axis TTE view or transgastric mid-papillary short-axis TEE view.

Finally, all MCS devices require appropriate positioning for optimal function, and some devices can be repositioned using bedside echocardiographic guidance. TTE can be used to evaluate inflow cannula position for VADs in a variety of views and to confirm laminar blood flow (with low peak velocity) using color-flow and continuous-wave Doppler. IABP position is generally not confirmed using TTE, as visualizing

the aortic-arch and descending thoracic aorta is challenging in most adults. However, TEE can be used to confirm or adjust IABP position. When correctly positioned, the IABP tip should be located approximately 2 to 3 cm below the left subclavian artery.

Other MCS devices for which echocardiography can be used to confirm correct position include Impellas, the TandemHeart VAD, and percutaneous RVAD-venovenous (VV) ECMO cannulas. The venous drainage cannula position in VA ECMO patients may be evaluated using TTE in the subxyphoid IVC view, but there is not much advantage over chest x-ray, except for evaluating IVC collapsibility around the cannula and thrombus on the cannula tip.

Catheter-Mounted Continuous Axial-Flow Ventricular Assist Devices (Impellas)

LV Impellas require their inflow to be positioned in the LV and their outflow to be positioned distal to the aortic valve in the ascending aorta. For the Impella RP, the inflow is positioned in the IVC and the outflow is positioned distal to the pulmonic valve in the PA. Assessment of device position usually begins with TTE parasternal views. For LV Impellas, the TTE parasternal long-axis view is a good starting point. This view allows both the inflow and outflow to be imaged simultaneously. For the Impella RP, the TTE parasternal short-axis RVOT view can be used. For Impellas, the echocardiographer must be careful to distinguish between the VAD itself and the pigtail.

For LV Impellas (2.5, 5.0, CP), the inflow should be positioned approximately 3.5 cm below the aortic valve. For the Impella 5.5, the inflow should be approximately 5 cm below the aortic valve. The measurement from the aortic valve to the inflow is generally obtained using a TTE parasternal long-axis view (Fig. 35.8). Importantly, the measurement does not include the pigtail, but is made from the aortic valve annulus to the inflow. Color-flow Doppler can be used to confirm appropriate positioning of the device's outflow in the ascending aorta. Normal turbulent blood flow should be visible above the aortic valve. If there is turbulent blood flow in the LVOT, then the outflow is not positioned correctly. Position can also be assessed using the motor current on the console, which should be pulsatile, and by examining either the differential pressure sensor (5.0 and RP) or the aortic waveform (2.5, CP, 5.5). For LV Impellas, the pigtail (if present) should not be entangled in the submitral apparatus and the inflow should not apply suction to the anterior leaflet of the mitral valve or significant MR can occur. If the pigtail is entangled in the submitral apparatus, rotation of the device should be attempted.³² For the Impella RV, the outflow should be 2 to 4 cm past the pulmonic valve with the pigtail sitting in the left or right PA.

Percutaneous Right Ventricular Assist Device-Venovenous Extracorporeal Membrane Oxygenation Cannulas (RA-PA Cannulas)

There are several commercially available percutaneous RVAD-VV ECMO cannulas including the ProtekDuo (29, 31, and 31 Fr Rapid Deployment) (TandemLife, Pittsburgh, PA) and the dual-lumen RV to PA cannula (24 Fr, 27 Fr, 31 Fr) (Spectrum Medical, Gloucester, England). For the ProtekDuo, the 29-Fr and 31-Fr catheters are used as percutaneous RVADs, while the rapid deployment catheter is used primarily for VV ECMO. When RVAD-VV ECMO cannulas are properly positioned, the inflow is positioned in the RA and the outflow is positioned in the main PA. The body of the cannula traverses the tricuspid valve and passes through the RVOT similar to a PA catheter. The outflow can be viewed with TTE or TEE using the TTE parasternal short-axis PA view or the TEE upper esophageal aortic-arch short-axis view. When properly positioned, the outflow orifices are usually 3 to 4 cm past the pulmonic valve and 1 to 2 cm proximal to the PA bifurcation (Fig. 35.9). Care should be taken to ensure that blood flow is not directed preferentially to the left or right PA or unilateral pulmonary edema can occur.

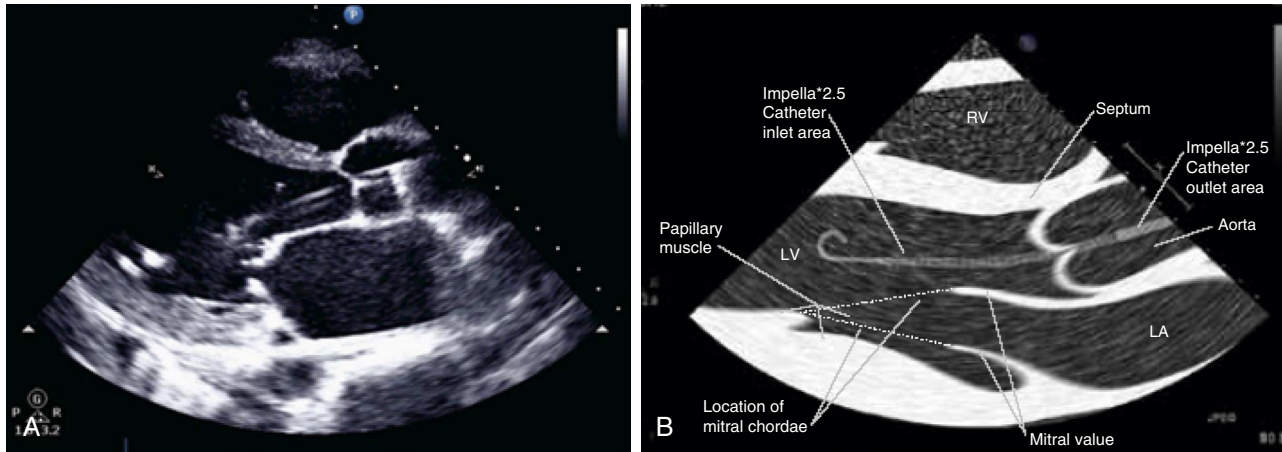


Figure 35.8 Transthoracic echocardiographic parasternal long-axis image of the Impella 2.5. Panel A shows the echocardiographic image and Panel B shows corresponding drawing with anatomy. The device inflow is positioned approximately 3.5 cm below the aortic valve annulus. The pigtail is also free from the subvalvular apparatus of the mitral valve. LA, Left atrium; LV, left ventricle; RV, right ventricle. (From Burzotta F, Trani C, Doshi SN, et al. *Impella ventricular support in clinical practice: collaborative viewpoint from a European expert user group*. *Int J Cardiol*. 2015;201:684-691.)

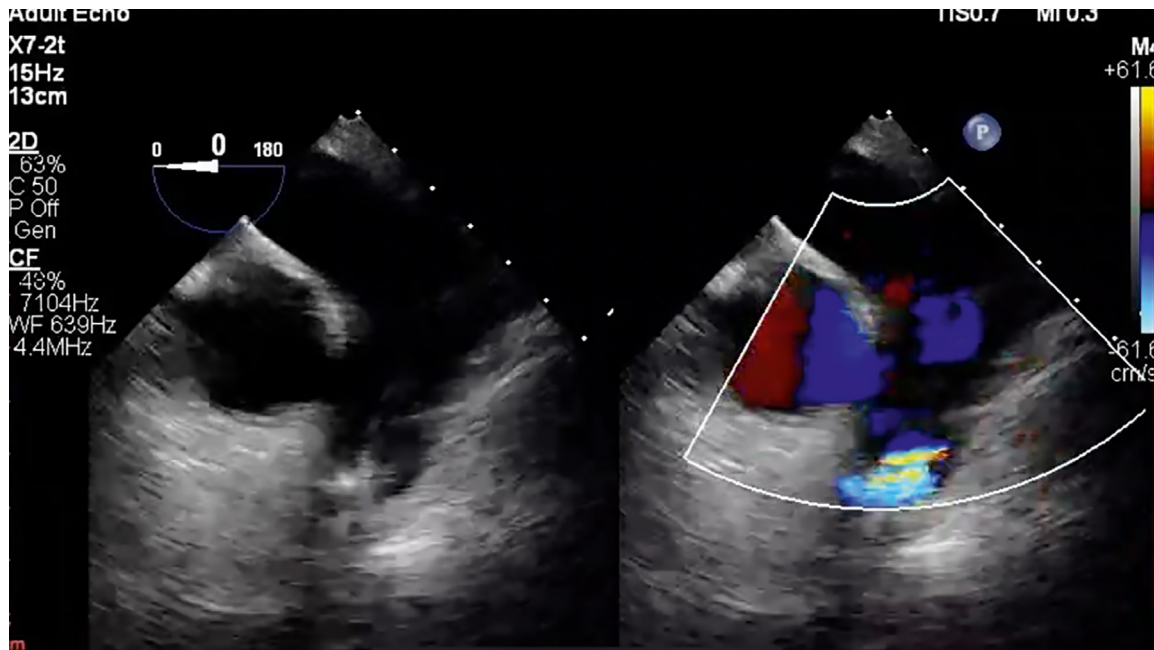


Figure 35.9 Transesophageal echocardiographic image (upper esophageal aortic-arch short-axis view) shows the ProtekDuo dual-lumen cannula positioned correctly in the main pulmonary artery.

Percutaneous Left Ventricular Assist Device (TandemHeart)

The TandemHeart drainage cannula (device inflow) crosses the atrial septum to drain blood from the LA. The drainage cannula holes should be entirely within the LA, as displacement toward the RA may draw deoxygenated blood into the system, creating shunt and hypoxemia when blood is returned into an artery.³³ Excessive pump speed can collapse the LA creating chatter and encouraging cannula migration toward the RA. Correct position can be confirmed using the TTE apical four-chamber or subxyphoid views where the cannula should be visualized crossing the atrial septum.

Total Artificial Heart

Imaging of the total artificial heart should begin with a TTE apical four-chamber view or TEE mid-esophageal four-chamber view. The device's air

diaphragms have an appearance that is similar to air in the LV during de-airing from cardiopulmonary bypass (CPB), making it difficult to visualize the two artificial ventricular chambers.³⁴ The device's Medtronic-Hall valves in the tricuspid and mitral positions can be imaged and a dominant central regurgitant jet is normal for these valves with one or two peripheral jets also seen around the tilting disc's border.³⁴ The valves in the pulmonic and aortic positions have a similar appearance and should be assessed qualitatively for normal opening and closing.

Hemodynamic Goals

General Principles

The principal hemodynamic goal in cardiogenic shock is to restore organ perfusion, by increasing total CO (intrinsic CO + MCS output). This will increase MAP if systemic vascular resistance (SVR) remains

TABLE 35.2 Different Types of Cardiogenic Shock and Their Features

Hemodynamic Variable	Preshock Normotensive Hypoperfusion	Preshock Hypotensive Normoperfusion	LV Dominant Shock	RV Dominant Shock	BiV Shock
Systolic arterial pressure, mm Hg	>90	<90	<90	<90	<90
CVP, mm Hg	Variable	Variable	<14	>14	>14
PCWP, mm Hg	Variable	Variable	>18	<18	Variable
CVP/PCWP	Depends on degree of LV and RV involvement	Depends on degree of LV and RV involvement	<0.86	>0.86	>0.86
PAPi (PAS-PAD)/RA	Depends on degree of RV involvement	Depends on degree of RV involvement	>1.5	<1.5 ^a	<1.5
Cardiac index, L/min/m ²	<2.2	≥2.2	<2.2	<2.2	<2.2
SVR, dynes s/cm ⁻⁵	>1600	800–1600	800–1600	800–1600	800–1600

^aRight ventricular (RV) dominant shock due primarily to RV dysfunction.

BiV, Biventricular; CVP, central venous pressure; LV, left ventricular; PAD, pulmonary artery diastolic pressure; PAPi, pulmonary artery pulsatility index; PAS, pulmonary artery systolic pressure; PCWP, pulmonary capillary wedge pressure; RA, right atrial pressure; SVR, systemic vascular resistance.

From Abraham J, Blumer V, Burkhoff D, et al. Heart failure-related cardiogenic shock: pathophysiology, evaluation and management considerations: review of heart failure-related cardiogenic shock. *J Cardiac Fail.* 2021;27:1126–1140.

constant. The formulas summarize the relationships between MAP, CO, and SVR, and between organ perfusion pressure and MAP.

$$\text{MAP} = \text{Total CO (intrinsic CO + MCS output)} \times \text{SVR}$$

$$\text{Organ perfusion pressure} = \text{MAP} - \text{higher of CVP or compartmental pressure (eg, intraabdominal pressure)}$$

When evaluating the adequacy of perfusion, both clinical and laboratory parameters should be used.³⁵ Specific indicators of inadequate tissue perfusion include cool extremities, declining mental status, weak pulses, increased capillary refill time (>3 s), urine output less than 0.5 mL/kg/h, lactate > 4 mmol/L, Svo₂ < 60%, and increasing transaminases or creatinine.

When tissue hypoperfusion occurs, a specific characterization of cardiogenic shock as isolated RV, LV, or biventricular failure is warranted (Table 35.2). Hemodynamic management is tailored to the disease state using a comprehensive approach that optimizes preload, afterload, heart rate, and contractility. Table 35.3 summarizes interventions that can be used to optimize each component of myocardial performance. In patients on MCS, device flow should be titrated with consideration of native CO to achieve a cardiac index (CI) > 2.2 L/min/m². Additional resuscitation targets include normalization of lactate (<2 mmol/L), Svo₂ > 65%, and stabilization of liver and kidney function when possible. In many patients, 3 to 4 L/min of MCS blood flow is adequate to augment CO, unload the RV or LV, and normalize perfusion. Arterial pulsatility (pulse pressure > 10 mm Hg) should be maintained in both the pulmonary and systemic circulations to prevent intracardiac thrombosis. If necessary, inotropes such as epinephrine, dobutamine, isoproterenol, or milrinone should be used. Inotrope selection is based on an individual patient's heart rate, MAP, and arrhythmia potential. Inhaled pulmonary vasodilators (e.g., inhaled nitric oxide, prostacyclin, and milrinone) are an important adjunct in treating RV failure, as RV stroke volume is highly sensitive to increased afterload.

For most organs, a MAP ≥ 65 mm Hg achieves adequate organ/tissue perfusion. However, this assumes a normal CVP and compartment pressure (e.g., intraabdominal pressure). Venous congestion, long-standing systemic arterial hypertension, intraabdominal hypertension, and intracranial hypertension are examples of states where a higher MAP may be needed to maintain adequate perfusion. There is significant evidence that MAP < 65 mm Hg is associated with organ failure in critically ill patients and most cardiogenic shock patients should have a MAP ≥ 65 mm Hg.^{36–38}

Unique Consideration for Intra-aortic Balloon Pump Patients

Correct timing/triggering of balloon inflation is critical for an IABP to achieve its hemodynamic benefits. Inflation can be triggered off

the arterial waveform, electrocardiogram, pacing impulse, automatic mode, or using the console's "internal" mode, which inflates the IABP at a fixed time interval (used during CPB). In patients with atrial fibrillation, the arterial waveform should be used to trigger IABP inflation. Incorrect triggering can cause deleterious hemodynamic effects or suboptimal IABP augmentation. For example, early inflation and late deflation cause the IABP to be up during systole, which increases cardiac afterload. Early deflation and late inflation lead to suboptimal IABP augmentation during diastole, which does not allow the patient to receive its full benefits. If IABP inflation and deflation timing is found to be incorrect on the monitor, it can be manually adjusted using controls on the console.

There is no standardized methodology for weaning an IABP, but generally it is switched from triggering on every diastole (1:1) to triggering on every other diastole (1:2). If this is well tolerated, a brief trial of every third diastole is sometimes performed. The IABP can be removed at the bedside with or without a vascular closure device. If a vascular closure device is not used, manual pressure should be held for at least 30 minutes after removal. Some centers use a femoral artery compression system such as the FemoStop (Abbott Cardiovascular, Plymouth, MN, USA) to limit blood loss and reduce the likelihood of developing a retroperitoneal hematoma. Various closure devices can be used including the Angio-Seal device (Terumo, Somerset, NJ, USA).

Unique Considerations for Left Ventricular Assist Device Patients

Hemodynamic management in an LVAD patient is focused mainly on optimizing RV function, setting optimal LVAD speed, and maintaining an appropriate MAP for perfusion. LVAD blood flow is proportional to the head pressure (inflow cannula pressure–outflow cannula pressure).³⁹ When the head pressure increases, blood flow through the LVAD decreases (Fig. 35.10). Hence blood flow through the LVAD is at its maximum during systole and its minimum during diastole. Treatments that improve RV function in patients with LVADs include inotropes (e.g., dobutamine, epinephrine, isoproterenol), inhaled pulmonary vasodilators (inhaled nitric oxide and prostacyclin), and other measures that reduce pulmonary vascular resistance (e.g., normothermia, avoidance of hypoxemia and acidosis) (see Chapter 22). Maintenance of a high MAP (80–90 mm Hg) was historically thought to optimize RV function, but recent studies have not confirmed this.⁴⁰ For most LVAD patients a MAP of 65 to 80 mm Hg is adequate.

For the HeartMate 3 LVAD, parameters that are continuously displayed on the monitor include estimated blood flow, pump speed in RPM, power in Watts, and pulsatility index (PI = power max – power min/power average). The normal range for PI is 3 to 5 and the number can be thought of as a general reflection of the amount of work

TABLE
35.3

Different Interventions for Optimizing Preload, Afterload, Contractility, and Heart Rate

	Preload	Afterload	Contractility	Heart Rate
Target	- Optimize preload when hypovolemic - Reduce congestion when volume overloaded RA 4–10 mm Hg, PCWP 10–18 mm Hg	- Optimize to maintain end-organ and tissue perfusion - Reduce excessive resistance to ventricular emptying MAP 60–80 mm Hg, SBP 80–140 mm Hg, SVR 900–1600 mm Hg	Achieve adequate end-organ and tissue perfusion with least increase to myocardial oxygen demand End-organ perfusion: lactate < 2, normalize renal and hepatic function	Optimize diastolic filling time to optimize left ventricular end-diastolic pressure HR 50–110 bpm
Increase	Intravascular volume - Careful fluid bolus or infusion - Blood transfusion Venous return - Passive leg raise - Compressive therapy - Ambulation/mobility	Vasopressors - Norepinephrine - Mixed α, β1, and β2 agonist ($\alpha > \beta$) 0.01–1 μg/kg/min - Dopamine - Lower dose: mild β1 agonist - Midrange dose: α, β1, and β2 agonist - Higher dose: increased α and β1 agonist 0.5–20 μg/kg/min - Epinephrine - Mixed α, β1, and β2 agonist 0.01–0.5 μg/kg/min - Phenylephrine a agonist 0.1–10 μg/kg/min - Vasopressin - V1 receptor in vascular smooth muscle 0.02–0.04 U/min	Inodilators - Phosphodiesterase-3 inhibitors - Milrinone 0.125–0.5 μg/kg/min - Levosimendan 0.05–0.2 μg/kg/min β-agonists - Dobutamine α, β1, and β2 agonist 2.5–20 μg/kg/min	b-agonists - Dopamine 0.5–20 μg/kg/min - Dobutamine 2.5–20 μg/kg/min - Epinephrine 0.01–0.5 μg/kg/min - Isoproterenol 2.0–20 μg/min (Withdrawing β-blockers) - Vagolytic - Atropine 0.5 mg IV q3–5 min - Max dose 3 mg - Pacing
Decrease	Decrease intravascular volume - Diuresis - Dialysis/CRRT Decrease venous return - Nitroglycerin 10–200 μg/min - Morphine - Positive pressure ventilation	Peripheral vasodilation - Nitroprusside 0.5–10 μg/kg/min - Milrinone 0.125–0.5 μg/kg/min - RASS inhibitors - ACE-I - ARB - ARNI - Positive pressure ventilation	Sedatives/anesthesia medications - Propofol 5–50 μg/kg/min - Dexmedetomidine 0.2–1.4 μg/kg/min - Prolonged tachycardia	Pharmacologic IV Amiodarone load: 150 mg IV $\times$ 1 over 1 min THEN 1 mg/min IV $\times$ 6 h, then 0.5 mg/min IV $\times$ 18 h Sedation/anxiolytic - Propofol - Dexmedetomidine DCCV if appropriate AF with RVR, VT with pulse
Special considerations	- Use PA catheter guidance whenever possible to assess RA and PCWP - Cautious in preload-sensitive states (e.g., mitral stenosis)	- Use PA catheter guidance whenever possible to assess SVR and PVR - Initiate GDMT when euvoletic and normotensive - Aortic stenosis - Afterload sensitive - Mitral regurgitation—consider afterload reduction	- Withdrawal of GDMT with β-agonist - Milrinone does not require cessation of β-blockers - Ventricular arrhythmias - Avoid increasing contractility in LVOT obstruction	Aortic regurgitation—consider maintaining elevated HR

ACE-I, Angiotensin-converting enzyme inhibitors; AF, atrial fibrillation; ARB, angiotensin receptor blockers; ARNI, angiotensin receptor neprilysin inhibitor; CCB, calcium channel blocker; CRRT, continuous renal replacement therapy; DCCV, direct current cardioversion; GDMT, guideline-directed medical therapy; HR, heart rate; LVOT, left ventricular outflow tract; MAP, mean arterial pressure; PA, pulmonary artery; PCWP, pulmonary capillary wedge pressure; RA, right atrial pressure; RASS, renin-angiotensin-aldosterone system; RVR, rapid ventricular response; SBP, systolic blood pressure; SVR, systemic vascular resistance; VT, ventricular tachycardia.

From Abraham J, Blumer V, Burkoff D, et al. Heart failure-related cardiogenic shock: pathophysiology, evaluation and management considerations: review of heart failure-related cardiogenic shock. *J Cardiac Fail.* 2021;27:1126–1140.

performed by the native heart in comparison to the LVAD. When the PI is high, the native heart is performing more work. Some conditions that lead to a high PI on the HeartMate 3 include hypertension and hypovolemia. Low PI can be from AI or inflow cannula obstruction. When PI drops on the HeartMate 3, it has a speed auto-reduction feature that leads to a drop in pump speed (typically 200–400 RPM below the set speed). Optimal LVAD speed should be determined using TTE or TEE, hemodynamic measurements, and biomarkers of tissue/organ perfusion. Increasing pump speed too much can create LV suckdown, reducing RV stroke volume and lowering CO.³⁰

Hemodynamic management for catheter-mounted axial continuous-flow VADs (Impellas) is similar, although the relationship between the head pressure and LVAD flow is slightly different, with a steeper inverse curve. Impellas do not display PI numerically, although the motor current is displayed and is pulsatile similar to that of the HeartMate 3.

Unique Considerations for Right Ventricular Assist Device Patients

RVADs have a similar relationship between head pressure and VAD flow. RVAD blood flow is highest during systole when RV and PA

pressures are similar and is lowest during diastole when the two pressures have a larger differential. Hemodynamic optimization of patients with an RVAD and isolated RV failure includes reducing pulmonary vascular resistance with inhaled pulmonary vasodilators and other previously described measures (e.g., normothermia, avoidance of hypoxemia and acidosis). Targeting euvoemia is critical in patients with RV failure. Determining volume status can be challenging and is usually based on multiple measures (CVP, natriuretic peptide levels, echocardiographic assessments of RV function, and Svo₂). In patients with an isolated RVAD and LV dysfunction, it is important to not have excessive blood flow to the LV, or the result can be pulmonary edema or hemorrhage.

Unique Considerations for Total Artificial Heart Patients

The total artificial heart has a maximum stroke volume of 70 mL from each artificial ventricle and can provide up to 9 L/min of CO. The usual stroke volume is closer to 50 to 65 mL. There is no concern about arrhythmias with the device because it provides biventricular support irrespective of cardiac rhythm. Typical postoperative settings include a LV drive pressure of 180 to 200 mm Hg (normal range 170–210 mm

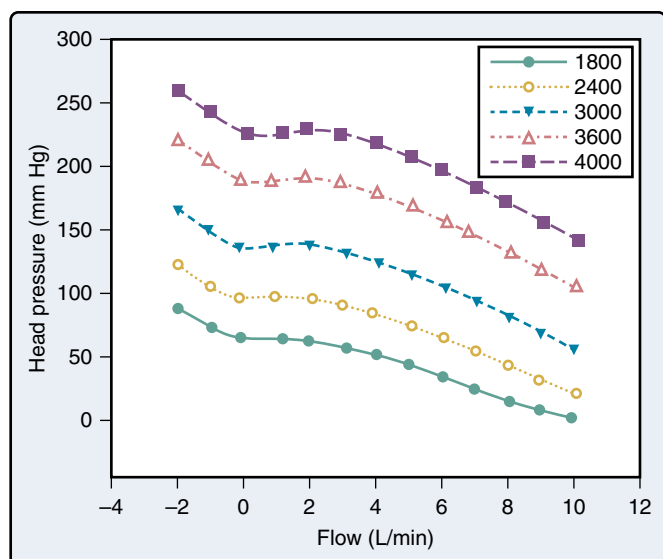


Figure 35.10 Relationship between head pressure (outflow pressure – inflow pressure) and ventricular assist device flow. (Redrawn from Moazami N, Fukamachi K, Kobayashi M, et al. Axial and centrifugal continuous-flow rotary pumps: a translation from pump mechanics to clinical practice. *J Heart Lung Transplant*. 2013;32:1-11.)

Hg), RV drive pressure of 20 to 30 mm Hg above the PA pressure (normal range 60–100 mm Hg), heart rate of 100 to 120, and 50% systole (normal range 50%–60%).²⁴ Vacuum pressure should not be applied until the sternum is closed (typically start at 10 mm Hg and do not exceed 30 mm Hg).²⁴ The device's manufacturer recommends using vasopressors to maintain a “normal” MAP and the driving pressure of the artificial ventricles should not be adjusted in order to achieve target MAP. Also, relatively low hemoglobin concentrations can be tolerated (e.g., 6 g/dL) because oxygen consumption is reduced with removal of the native cardiac ventricles.

Unique Considerations for Venoarterial Extracorporeal Membrane Oxygenation Patients

In VA ECMO patients, total CO is equal to native CO plus ECMO circuit blood flow (more detail covered in [Chapter 27](#)). In most patients, 3 to 4.5 L/min of ECMO circuit blood flow is enough to improve perfusion, normalize hemodynamics, and eliminate shock. The optimal MAP during VA ECMO is not known, but observational studies suggest that lower MAPs are associated with worse outcome.⁴¹ Alternatively, maintenance of a high MAP may prevent cardiac ejection in patients with severely impaired LV function. In most patients, a pulse pressure ≥ 10 mm Hg should be targeted and MAP should be between 65 and 80 mm Hg. LV venting has been increasingly recognized as an important adjunct in peripheral VA ECMO because peripheral VA ECMO increases afterload and does not reduce preload to the same degree as central VA ECMO. LV venting is associated with improved survival in multiple observational studies.⁴² LV venting strategies include use of an IABP, surgical LA or LV venting, use of a catheter-mounted axial continuous-flow LVAD, or atrial septostomy.⁴³

Anticoagulation for Mechanical Circulatory Support Devices

Intra-aortic Balloon Pump

Routine systemic anticoagulation is not required if an IABP triggers 1:1.⁴⁴ In fact, routine systemic anticoagulation is associated with more bleeding and does not reduce thrombotic complications ([Box 35.4](#)). In a randomized controlled trial that included over 150 IABP patients, heparinization with a goal activated partial thromboplastin time (aPTT)



BOX 35.4 ANTICOAGULATION FOR MCS

- IABP does not require routine anticoagulation
- Venoarterial extracorporeal membrane oxygenation (ECMO) patients are usually anticoagulated with heparin, although optimal anticoagulation intensity remains controversial, and some centers have reported good outcomes with no anticoagulation
- Direct thrombin inhibitors have increasingly been used in venoarterial ECMO patients, but there is no strong evidence to support their superiority as first-line agents
- Both TandemHeart and Impella percutaneous VADs utilize a purge solution, which usually contains heparin, and low-dose systemic anticoagulation is also commonly administered to obtain an activated partial thromboplastin time (aPTT) of two to three times normal
- For critically ill patients with paracorporeal and durable VADs heparin is usually used as a first-line anticoagulant and in many centers aspirin is also administered
- The 4T score can be used to risk stratify MCS patients for heparin-induced thrombocytopenia (HIT)
- Direct thrombin inhibitors (e.g., bivalirudin and argatroban) remain the first-line agents for treating HIT

IABP, Intraaortic balloon pump; MCS, mechanical circulatory support; VADs, ventricular assist devices.

of 50 to 70 seconds was associated with a sevenfold increase in bleeding (14.1% vs 2.4%) and no reduction in thrombotic complications.⁴⁵

Venoarterial Extracorporeal Membrane Oxygenation

Anticoagulation for VA ECMO is covered in detail in [Chapter 27](#).

Percutaneous Ventricular Assist Devices

Catheter-mounted axial continuous-flow VADs (Impellas) have a “purge” solution which is standardly composed of dextrose in water and 25 to 50 international units (IU) per mL of heparin. The most common concentration of heparin used in the purge solution is 50 IU/mL, but 25 IU/mL also appears to be acceptable in most cases.⁴⁶ The purge solution prevents blood from entering the motor and thromboembolic stroke. Typically, the purge solution runs near 8 to 20 mL/h and this leads to a mild elevation of the aPTT (~40–50 s). In patients with heparin-induced thrombocytopenia (HIT), alternative strategies have been used for anticoagulation including dextrose only purge solution combined with systemic administration of a direct thrombin inhibitor (DTI) or adding a DTI to the purge solution directly.^{47–49} Impella's manufacturer recommends a target activated clotting time (ACT) of 160 to 180 seconds, but they do not provide specific recommendations for aPTT. An aPTT goal of 55 to 80 seconds has been recommended by experts.³² The TandemHeart percutaneous LVAD has an internal purge solution. The device's manufacturer recommends that a solution containing 45,000 IU of heparin in 500 mL of normal saline be infused at 10 mL/h.⁵⁰ In addition to this, patients benefit from systemic heparinization with a goal aPTT of two to three times normal (e.g., 55–75 s).⁵⁰ Dual-lumen RVAD-VV ECMO cannulas have no standardly accepted anticoagulation regiment, but an aPTT in the range of 45 to 55 seconds is reasonable, as this is commonly used for other VV ECMO patients, and there is minimal risk for left-sided thromboembolism in patients who do not have an intracardiac shunt (e.g., patent foramen ovale).

Total Artificial Heart and Paracorporeal Ventricular Assist Devices

The anticoagulation approach for the total artificial heart differs by institution, but most published approaches use unfractionated heparin,

low-dose aspirin (81 mg), dipyridamole (50 mg q8h), and pentoxifylline. Anticoagulation is started when postoperative bleeding slows. Minimal chest drain output (<50 mL/h for 3 consecutive hours), normovolemia, and a stable hemoglobin are signs that anticoagulation can be safely initiated. Eventually, heparin can be switched for warfarin in patients who are stable and do not have an anticipated need for holding anticoagulation.⁵¹ Bridging with heparin should be performed until the patient has an international normalized ratio (INR) between 2 and 3.

There are many iterations of paracorporeal VAD support for isolated LV, RV, or biventricular failure and anticoagulation regimens differ by pump. The CentriMag heart pump is fully magnetically levitated and has no bearings or seals, reducing the rate of pump thrombosis. CentriMag's manufacturer recommends that systemic anticoagulation be given, but no specific target ACT or aPTT is recommended. A published report from a high-volume center described using an aPTT goal of 60 to 80 seconds or ACT of 160 to 180 seconds in combination with 81 mg of aspirin.⁵² With this protocol, there was a relatively high bleeding rate near 50%.⁵² A second multicenter trial targeted an ACT of 200 seconds and reported a much lower bleeding rate of 5%.⁵³ When used for very short-term support (<14 days), excellent results have been reported for the CentriMag pump without any systemic anticoagulation.⁵⁴

Durable Left Ventricular Assist Devices

Anticoagulation for the HeartMate 3 is similar to that for paracorporeal VADs. Postsurgical bleeding should be minimal, as is evidenced by normovolemia, a stable hemoglobin, and minimal chest drain output (<50 mL/h for 3 consecutive hours). Intravenous unfractionated heparin is started with a low initial aPTT goal (e.g., 45–55 s) and is titrated up to 60 to 80 seconds 48 to 72 hours after surgery.⁵⁵ The 2013 guidelines from the International Society of Heart and Lung Transplantation recommend the use of aspirin with systemic anticoagulation.⁵⁶ Usually aspirin is started on postoperative day 2 or 3.⁵⁵ Warfarin can be started when the patient is stabilized and there is no anticipated need to withhold anticoagulation. Bridging with heparin should be performed until the patient has an INR between 2 and 3.

Heparin-Induced Thrombocytopenia Evaluation and Treatment

HIT evaluation for MCS patients should be similar to that in non-MCS patients. The 4T score can be used for risk stratification and studies have suggested that it performs similarly well in patients on MCS.⁵⁷ In general, patients with an optical density < 1 unit on HIT antibody enzyme linked immunosorbent assay (ELISA) are unlikely to have true HIT, whereas the likelihood is high (>90%) in patients who have an optical density > 2 units.⁵⁸ In a patient with an intermediate or high probability 4T score and a high optical density > 1 unit, heparin should be stopped and a DTI should be administered. Bivalirudin has a short half-life of approximately 20 to 30 minutes and is mainly cleared in an organ-independent manner (proteolytic cleavage). However, its half-life is prolonged up to 2 hours in patients with significant chronic kidney disease or acute renal failure.⁵⁹ Argatroban has a slightly longer half-life (40–50 minutes), is principally cleared through hepatic mechanisms, and should not be used in patients with cirrhosis or acute hepatic failure.⁵⁹ Bivalirudin can be started at a dose of 0.1 mg/kg/h in critically ill patients with organ dysfunction and suspected HIT; whereas, the starting dose for argatroban is between 0.25 and 0.5 µg/kg/min.^{59,60}

Infection Surveillance and Prevention

Cardiogenic shock patients have impaired immunity and are at high risk for health care–associated infections (HAIs) including blood stream infections (BSIs), surgical site infections (SSIs), and ventilator-associated pneumonia (VAP).^{61,62} There are multiple risk factors for



BOX 35.5 INFECTION SURVEILLANCE AND PREVENTION

- Mechanical circulatory support (MCS) patients are at high risk for health care–associated infections
- There is no strong evidence to support the use of prophylactic antibiotics in patients on temporary MCS
- Cultures should only be performed when there is moderate-to-high suspicion for infection and “pan culturing” and “surveillance culturing” are not appropriate in patients on temporary MCS
- When there is moderate-to-high suspicion for infection, broad-spectrum antibiotics should be started, targeted cultures should be performed, and antibiotics narrowed after appropriate sensitivities become available

HAIs in this group of patients including large vascular access catheters, indwelling urinary catheters for hourly monitoring of urinary output, and invasive mechanical ventilation for acute hypoxemic respiratory failure. For this reason, MCS patients deserve heightened surveillance and it is critical to follow Centers for Disease Control and Prevention (CDC) practices for HAI prevention in the ICU (Box 35.5).

In an observational study that included 220 VA ECMO patients with cardiogenic shock, the incidence of nosocomial infection was 64% and patients who developed septic shock had a twofold increase in mortality.⁶¹ In a second analysis that specifically examined the risk of BSI during VA ECMO, the incidence was 8 BSIs per 1000 ECMO days with 5.7% of patients developing infection.⁶³ An observational study that included over 10,000 patients with durable MCS who were in the International Society of Heart and Lung Transplantation Mechanically Assisted Circulation Support Registry found that 12% of patients developed a BSI with the highest incidence in the first 3 months after implantation and risk falling by 50% over the next 3 months.⁶⁴ The incidence of BSI in patients with temporary percutaneous LVADs is not well described, but one study of the TandemHeart LVAD reported an infection rate of 16%.⁶⁵

There is no evidence to support the use of prophylactic antibiotics in patients who are placed on MCS. In fact, reported infection rates are more or less the same in the published literature, regardless of whether prophylactic antibiotics are administered.⁶⁶ Patients with delayed sternal closure are at high risk for infection, pneumonia in particular, but there is no high-level evidence to support prolonged prophylactic antibiotics in this population and increasing the duration of prophylaxis beyond 24 hours may cause harm.^{67–69} Surveillance cultures are not recommended for MCS patients, but targeted cultures should be sent when there are clinical signs of infection (e.g., fever or hypothermia, new hypotension or tachycardia, worsening mental status, new organ failure, rigors, changes in secretions, new chest infiltrates on imaging, or leukocytosis or leukopenia). In some MCS patients, particularly those on continuous renal replacement therapy or ECMO, fever may be masked by blood cooling in the circuit. When evaluating for BSI, peripheral blood cultures should be sent rather than cultures from an existing arterial or central venous catheter.

Hemolysis, Transfusion, and Hemostasis Management

Hemolysis

Severe hemolysis related to red blood cell (RBC) trauma can cause acute anemia and renal failure in MCS patients. Hemolysis risk increases in patients with circuit, oxygenator, or pump thrombosis, and also when circuit RPM are increased. Cell-free hemoglobin dimers are pro-oxidant, enhancing lipid peroxidation of arachidonic acid, and subsequent formation of prostaglandins, F₂-isoprostanes, and isofurans.⁷⁰ Increased concentrations of these molecules are associated with

the development of renal injury through vasoconstriction and direct renal toxicity in the proximal tubule.^{70,71} Cell-free hemoglobin also depletes physiologic nitric oxide stores, which are needed to maintain normal vasodilation.⁷²

In VA ECMO patients, peak plasma-free hemoglobin concentration is an independent risk factor for acute renal failure.⁷³ Severe hemolysis (50 mg/dL or 500 g/L) is relatively uncommon during VA ECMO, occurring in less than 5% of patients.⁷⁴ Between one-third and two-thirds of patients with catheter-mounted axial continuous-flow LVADs develop hemolysis and it is an independent risk factor for acute renal failure.^{75–77} Longer duration of support appears to be an important risk factor for increased hemolysis in these patients.⁷⁷ The CentriMag pump causes minimal hemolysis in in vitro simulation studies, but clinical data on the incidence of severe hemolysis are relatively sparse.^{78,79}

The two most common biomarkers used to monitor for hemolysis during MCS are lactate dehydrogenase (LDH) and plasma-free hemoglobin. LDH is more readily available and is run in-house at many large hospitals. LDH is nonspecific, is produced in almost all tissues, and can be elevated in multiple pathologic conditions including MI, acute liver injury, acute kidney injury, pancreatitis, and rhabdomyolysis. Plasma-free hemoglobin is a more specific biomarker for hemolysis, but in many hospitals turnaround time is 48 hours or longer. Point-of-care devices for measuring free hemoglobin exist and are approved for clinical use in the United States (e.g., HemoCue, Ängelholm Sweden). However, these devices have not yet reached widespread clinical use.⁸⁰ Hemoglobinuria should be suspected when a patient's urine changes to a pink color, but remains translucent. Both hematuria and hemoglobinuria give a positive dipstick value for blood, but when the urine is centrifuged patients with hematuria have a clear supernatant and those with hemoglobinuria have a pink supernatant.

Management of hemolysis in patients on MCS starts with a careful evaluation of the circuit, pump, and purge pressures for pertinent devices. When purge pressure is elevated and there are no kinks or obstructions in the tubing, it is strongly suggestive of device thrombosis. In VA ECMO patients, the cannulas and oxygenator should be inspected for thrombus with a flashlight. For any device, if there is severe hemolysis and corresponding evidence of circuit, oxygenator, or device thrombosis, consideration must be given to changing it. Patients with durable LVADs represent a unique case, as redo-sternotomy with LVAD replacement is not a trivial surgical procedure. When device exchange is not possible, and thrombosis is suspected, DTIs or low-dose thrombolytics may be used to reduce thrombus size within the LVAD. Thrombolysis is associated with an almost 30% rate of hemorrhagic complications when used for this indication.^{81,82} In some cases, hemolysis can be mitigated by turning down device RPM to the lowest tolerated level.

Few interventions have been shown to be effective in mitigating the nephrotoxic effects of free hemoglobin. Sodium bicarbonate infusion may reduce the incidence of acute kidney injury in patients who have hemolysis, but there are no randomized trials to support its use in MCS patients.^{83,84}

Transfusion and Hemostasis Management

Patients on MCS have a high incidence of anemia for multiple reasons including chronic comorbidities, acute renal failure, and hemolysis. At present, the standard of care for critically ill adults, including those in cardiogenic shock, is to transfuse RBCs to maintain a blood hemoglobin concentration greater than or equal to 7 g/dL (Box 35.6). Restrictive transfusion has been found to be safe in multiple randomized clinical trials and RBC transfusion is also associated with mortality in numerous patient populations including STEMI patients and VA ECMO patients.^{85,86} A recent meta-analysis of randomized controlled trials that compared liberal versus conservative RBC transfusion suggested that restrictive transfusion may be superior in terms of safety.⁸⁷

Hemostasis management in patients on MCS is challenging because the risks of procoagulant therapies are amplified. In any given situation



BOX 35.6 TRANSFUSION AND HEMOSTASIS MANAGEMENT

- A red blood cell transfusion trigger of 7 g/dL is appropriate for most non-bleeding mechanical circulatory support (MCS) patients
- There is no strong evidence to support specific platelet transfusion triggers in MCS patients, but platelet count probably should be above 50×10^9 per liter for most MCS patients
- Fibrinogen concentrate or cryoprecipitate can be used to treat hypofibrinogenemia (i.e., fibrinogen < 200 mg/dL) in bleeding MCS patients
- Potent procoagulant therapies (e.g., prothrombin complex concentrate, recombinant activated factor VII, and von Willebrand Factor concentrate) have been used to treat severe bleeding during MCS, but thrombosis risk and its consequences must be carefully considered prior to treatment

the risk of life-threatening thromboembolism must be weighed against the risks of continued bleeding and allogeneic transfusion. Exposure of blood to centrifugal and axial-flow blood pumps lead to supraphysiologic shear stress. This causes severe alterations in primary hemostasis because of increased platelet surface receptor shedding and loss of large von Willebrand Factor (vWF) multimers.^{88–92} Generally, hemostasis management can be in accordance with recent published national guidelines; however, some nuances require special consideration.^{93,94}

First, in patients with severe life-threatening bleeding anticoagulation should be held. The risks for holding anticoagulation differ by device, arterial cannulation scheme, and also based on the patient's clinical condition. In bleeding patients with hypofibrinogenemia (fibrinogen < 200 mg/dL) cryoprecipitate or fibrinogen concentrate should be given; if platelet count is less than 50×10^9 /L, platelet transfusion should be given; and if the count is between 50 and 100×10^9 /L platelet transfusion should be considered because of the qualitative platelet dysfunction that occurs during ECMO. If INR is greater than 1.5, fresh frozen plasma (FFP) transfusion should be considered, but each unit of FFP only raises coagulation factor levels by 3% to 5% in an average-sized adult. When these abnormalities have been corrected and coagulopathic bleeding persists, vWF concentrate treatment may be considered (40 IU/kg) if acquired von Willebrand Syndrome is thought to be a contributing factor to coagulopathic bleeding.⁹⁵

Finally, there are multiple case reports of using both recombinant activated factor VII and prothrombin complex concentrate (PCC) off-label in ECMO patients and other MCS patients.^{96,97} These agents should be considered high risk and administration should only occur after a multidisciplinary discussion. PCC may be used (on-label) to safely reverse warfarin effect in durable LVAD patients with life-threatening bleeding.^{98,99}

Arrhythmias, Cardiac Arrest, and Cardiopulmonary Resuscitation

Cardiac arrhythmias are common in MCS patients, particularly those with ischemic heart disease or chronic cardiomyopathies.¹⁰⁰ Depending on the device and MCS configuration, some arrhythmias are relatively well tolerated. Basic arrhythmia management includes addressing hemodynamic instability, identifying the arrhythmia's cause when possible, and making appropriate mechanical or pharmacological corrections.

With the exception of total artificial heart patients and other biventricular assist device (BiVAD) patients, most MCS patients rely on having some intrinsic cardiac function, which requires a stable cardiac rhythm.¹⁰¹ Even VA ECMO patients typically require some intrinsic cardiac function to meet their metabolic demands and prevent blood stagnation in the LV. When an unstable arrhythmia occurs in an MCS patient, advanced cardiac life support (ACLS) protocols should be

followed. Even if target MAP is maintained with MCS, ventricular tachyarrhythmias should be treated because they increase myocardial oxygen consumption and can cause LV distention. MCS devices are not a contraindication to cardioversion and patients with unstable arrhythmias should be cardioverted.¹⁰² Patients with chronic, advanced CHF, frequently have implantable cardiac defibrillators (ICDs), which can treat ventricular arrhythmias with antitachycardia pacing, cardioversion, or defibrillation.

Vasopressors (e.g., norepinephrine or vasopressin) can be used to temporarily increase MAP in MCS patients with hypotension from an arrhythmia until cardioversion is performed. If the patient has a BiVAD or is on VA ECMO, increasing pump speed and device output may help to empty a distended heart, increase MAP, and improve the likelihood of successful cardioversion. This strategy, however, will not work in a patient with an isolated RVAD because LV output will not be adequate with an unstable cardiac rhythm.¹⁰¹ In contrast, LVAD patients with normal pulmonary vascular resistance may be able to maintain a relatively normal MAP in the setting of ventricular arrhythmias because blood can be passively pushed through the pulmonary circulation by the RV if CVP is high. Pharmacologic treatment of arrhythmias in covered in detail in [Chapters 8 and 34](#).

Common causes of arrhythmias that can be rapidly assessed and treated include electrolyte and acid/base disturbances (e.g., acidemia). Device-specific problems should also be considered. For example, stimulation of the myocardium from ventricular suckdown should be ruled out. In the case of ventricular suckdown, turning down pump speed may help to normalize cardiac rhythm. Misplaced or migrated intracardiac devices, including PA catheters and percutaneous LVADs, can stimulate the myocardium directly leading to ventricular arrhythmias. A quick survey of device position at the skin to check for inadvertent migration is warranted. It is worth noting that durable VADs implanted through the myocardial wall are generally not arrhythmogenic when examined by electrophysiologic studies.¹⁰³

Refractory tachyarrhythmias require consultation from a cardiac electrophysiologist, particularly in a patient with extensive scarring from prior ischemic injury.¹⁰⁴ Even with anticoagulation, these patients may be at high risk for thromboembolic events due to regional blood stasis in the heart.¹⁰⁵ In some cases, an invasive ablation may be necessary and is possible on MCS.¹⁰⁶ Sympathectomy can be useful in treating catecholamine-induced electrical storm.¹⁰⁷ Bedside ultrasound-guided stellate ganglion block can be performed, and if successful, can be followed by thoracoscopic sympathectomy.¹⁰⁷ In patients with refractory atrial fibrillation with rapid ventricular response, AV nodal ablation with ventricular pacemaker implantation may be a last resort.¹⁰⁸ Bradyarrhythmias are less common in MCS patients but can occur in patients with right coronary artery disease or in cases of heart block from AV nodal injury. Most bradyarrhythmias are initially treated with chronotropic drugs (e.g., epinephrine, dobutamine, dopamine, or isoproterenol), anticholinergic agents (e.g., atropine), or temporary transvenous or epicardial pacing. Severe bradycardia should be avoided in MCS patients because it can reduce cardiac ejection and increase ventricular distention.

Chest compressions should be performed in isolated RV or LV MCS patients who have a cardiac arrest (pulseless electrical activity or ventricular arrhythmia) and have severe hypotension. Patients with a total artificial heart do not require chest compressions, as the device provides biventricular support and the artificial ventricles cannot be physically compressed. Similarly, VA ECMO patients and BiVAD patients have biventricular support and chest compressions are of limited utility if the devices are functioning properly and providing adequate blood flow. In patients with catheter-mounted axial continuous-flow VADs, there is a risk for device migration after chest compressions.¹⁰⁹ Hence, echocardiography should be performed to confirm correct device position after resuscitation. In cases of isolated RV or LV MCS and refractory cardiac arrest, consideration should be given to switching to a biventricular support strategy (e.g., VA ECMO).

Mobility, Physical Therapy, and Nutrition

Early mobilization and physical therapy continues to be an area of innovation in MCS care.¹¹⁰ Bed confinement leads to muscle weakness and atrophy, as well as pressure injuries, pulmonary complications, and development of a proinflammatory state, which propagates further loss of muscle protein.¹¹¹ There is growing evidence that the detrimental effects of immobility persist for many years following ICU discharge, with survivors experiencing prolonged impairments in functional capacity and cognitive function.¹¹² Early mobility programs have a clear benefit in critically ill patients, improving functional outcomes and decreasing the duration of mechanical ventilation, incidence of delirium, and hospital length of stay.^{113,114} In contemporary ICU practice, MCS patients are commonly extubated,¹¹⁵ and in many cases can participate in ambulation and physical therapy. Even patients on peripheral VA ECMO with femoral arterial cannulation can ambulate safely in experienced centers with minimal complication.^{116,117} In a recent systematic review that included over 300 ECMO patients, physical therapy was associated with a <1% rate of cannula complications and with improvements in functional capacity, muscle strength, and reduced hospital length of stay.¹¹⁸

It is extremely important for critically ill MCS patients to receive adequate nutrition. MCS patients who do not receive adequate nutrition have increased mortality and patients are at particularly high risk for being deprived of nutrition early in their MCS course, when their shock is resolving.¹¹⁹ The 2016 American Society for Parenteral and Enteral Nutrition (ASPEN) guidelines for provision of nutritional support in ICU patients recommend that all patients have their caloric and protein requirements calculated at ICU admission, have enteral nutrition started within 48 hours of ICU admission when possible, have steps taken to avoid aspiration (e.g., head of bed elevated 30–45 degrees, use of promotility agents, etc.), and not have gastric residual volumes measured to make decisions about continuing feeding.¹²⁰ Parenteral nutrition is not recommended for at least 7 days in patients with low caloric intake, but the quality of evidence is “very low” per the guideline.¹²¹

Triage Pathway for Patients on Temporary Mechanical Circulatory Support

Patients with cardiogenic shock on temporary MCS must be bridged to recovery, heart transplantation, or durable MCS (typically a durable LVAD) to survive. Although seemingly straightforward, this triage pathway is complex and has many nuances. For example, at the time of MCS initiation, some patients appear to be good candidates for bridging to heart transplantation, but over time they become poor candidates because of complications (e.g., infection, acute stroke). Alternatively, some patients who are thought to have a low likelihood of cardiac recovery are successfully weaned off MCS. Generally speaking, temporary MCS should be used in patients who are likely to recover adequate cardiac function or in patients who can be bridged to heart transplantation or durable LVAD ([Box 35.7](#)).

In 2016, the International Society of Heart and Lung Transplantation (ISHLT) published an update to their guideline on listing criteria for heart transplantation.¹²² Candidate patients should have irreversible cardiac failure, reasonably good functional status (ability to stand and ambulate), stable mental health, adequate nutritional status, and solid social support. Select patients with LV failure who cannot undergo heart transplantation and patients who have a long waiting time for heart transplantation can be converted from temporary MCS to a durable LVAD, either as destination therapy or as a bridge to transplant.

Patients who recover enough cardiac function to wean from temporary MCS but are left with significant CHF symptoms should be optimized using goal-directed medical therapy in accordance



BOX 35.7 TRIAGE PATHWAYS FOR PATIENTS ON TEMPORARY MCS

- Potential endpoints for temporary MCS patients include cardiac transplantation, durable LVAD implantation, and cardiac recovery
- Cardiac transplantation evaluation requires a multidisciplinary approach with multiple factors considered including age, immunological status, blood type, psychosocial history, body weight, substance abuse history, pulmonary vascular resistance, functional status, and other organ function

MCS, Mechanical circulatory support.



BOX 35.8 ETHICAL CONSIDERATIONS

- Early conversations outlining what temporary mechanical circulatory support (MCS) can and cannot accomplish are important for setting reasonable expectations with patients and their families
- Periodic reevaluation of patient's overall status and potential MCS endpoints should be performed
- State laws and hospital policies about surrogate decision-making should be referenced when making complex decisions about end-of-life care in incapacitated patients on MCS
- In cases of conflict or disagreement with patients or their family members, clinicians should attempt to understand their desires, culture, values, and build consensus before making complex medical decisions

with American Heart Association guidelines.¹²³ This includes use of β -blockers, renin-angiotensin system blocking drugs, and aldosterone inhibitors if tolerated. Use of these agents is associated with significant reductions in mortality, hospitalization, and improvements in LV ejection fraction.^{124,125}

Ethical Considerations

Temporary MCS creates complex ethical dilemmas, particularly when a “bridge-to-nowhere” scenario arises and the patient cannot be transplanted or have a durable LVAD implanted, and there is no reasonable likelihood of cardiac recovery. In some cases, this leads to a protracted MCS course of weeks to months. In patients with irreversible cardiac failure who cannot be transplanted, MCS becomes necessary for survival, and in some cases surrogates find it difficult to make a decision about terminating MCS, because of the threat of immediate cardiovascular collapse. In these cases, one option is to discuss/negotiate “nonescalation” of care, meaning that if the MCS circuit becomes thrombosed or compromised, it is not replaced. Similarly, limits can be set on adding “new” therapies that would be futile such as renal replacement therapy, broad-spectrum antibiotics, or high-dose vasopressors. For MCS patients, it is critical to have “up-front” conversations about what temporary MCS can and cannot accomplish and goals of care should frequently be revisited during an MCS run. Ideally, conversations should be multidisciplinary and should attempt to address any questions about MCS that the family has (Box 35.8).

In cases of conflict or disagreement between the family and health care team about terminating MCS it is best to give the family time to reconcile the medical facts, consider the patient's wishes, and to slowly build multidisciplinary consensus. Families have different styles/processes of grieving and coping, historical issues can impact their decision-making, and religious beliefs are important for many families.¹²⁶ Physicians and nurses should be sensitive to these complexities and try to be active listeners and work toward mutual solutions in the best interest of the patient.

REFERENCES

- Osman M, Syed M, Patibandla S, et al. Fifteen-year trends in incidence of cardiogenic shock hospitalization and in-hospital mortality in the United States. *J Am Heart Assoc.* 2021;10:e021061.
- Deshpande SR, Slepian MJ, Alsoufi B. HeartWare HVAD market withdrawal and impact on the pediatric field. *ASAIO J.* 2021;67:825–826.
- Levy B, Bastien O, Karim B, et al. Experts' recommendations for the management of adult patients with cardiogenic shock. *Ann Intensive Care.* 2015;5:52.
- Muir AL, Kirby BJ, King AJ, Miller HC. Mixed venous oxygen saturation in relation to cardiac output in myocardial infarction. *Br Med J.* 1970;4:276–278.
- Fincke R, Hochman JS, Lowe AM, et al. Cardiac power is the strongest hemodynamic correlate of mortality in cardiogenic shock: a report from the SHOCK trial registry. *J Am Coll Cardiol.* 2004;44:340–348.
- Grodin JL, Mullens W, Dupont M, et al. Prognostic role of cardiac power index in ambulatory patients with advanced heart failure. *Eur J Heart Fail.* 2015;17:689–696.
- Tongers J, Sieweke JT, Kuhn C, et al. Early escalation of mechanical circulatory support stabilizes and potentially rescues patients in refractory cardiogenic shock. *Circ Heart Fail.* 2020;13:e005853.
- Kantrowitz A, Tjonneland S, Freed PS, Phillips SJ, Butner AN, Sherman Jr JL. Initial clinical experience with intra-aortic balloon pumping in cardiogenic shock. *JAMA.* 1968;203:113–118.
- De Silva K, Lumley M, Kailey B, et al. Coronary and microvascular physiology during intra-aortic balloon counterpulsation. *JACC Cardiovasc Interv.* 2014;7:631–640.
- van Nunen LX, Noc M, Kapur NK, Patel MR, Perera D, Pijls NH. Usefulness of Intra-aortic balloon pump counterpulsation. *Am J Cardiol.* 2016;117:469–476.
- Schampaert S, van Nunen LX, Pijls NH, et al. Intra-aortic balloon pump support in the isolated beating porcine heart in nonischemic and ischemic pump failure. *Artif Organs.* 2015;39:931–938.
- Perera D, Stables R, Thomas M, et al. Elective intra-aortic balloon counterpulsation during high-risk percutaneous coronary intervention: a randomized controlled trial. *JAMA.* 2010;304:867–874.
- Patel MR, Smalling RW, Thiele H, et al. Intra-aortic balloon counterpulsation and infarct size in patients with acute anterior myocardial infarction without shock: the CRISP AMI randomized trial. *JAMA.* 2011;306:1329–1337.
- van Nunen LX, van 't Veer M, Schampaert S, et al. Intra-aortic balloon counterpulsation reduces mortality in large anterior myocardial infarction complicated by persistent ischemia: a CRISP-AMI substudy. *EuroIntervention.* 2015;11:286–292.
- Thiele H, Zeymer U, Neumann FJ, et al. Intra-aortic balloon counterpulsation in acute myocardial infarction complicated by cardiogenic shock (IABP-SHOCK II): final 12 month results of a randomised, open-label trial. *Lancet.* 2013;382:1638–1645.
- Thiele H, Zeymer U, Neumann FJ, et al. Intra-aortic balloon support for myocardial infarction with cardiogenic shock. *N Engl J Med.* 2012;367:1287–1296.
- Vargas KG, Jager B, Kaufmann CC, et al. Impella in cardiogenic shock following acute myocardial infarction: a systematic review and meta-analysis. *Wien Klin Wochenschr.* 2020;132:716–725.
- Elia E, Iannaccone M, D'Ascenzo F, et al. Short term outcomes of Impella circulatory support for high-risk percutaneous coronary intervention: a systematic review and meta-analysis. *Catheter Cardiovasc Interv.* 2022;99(1):27–36.
- Patel SM, Lipinski J, Al-Kindi SG, et al. Simultaneous venoarterial extracorporeal membrane oxygenation and percutaneous left ventricular decompression therapy with Impella is associated with improved outcomes in refractory cardiogenic shock. *ASAIO J.* 2019;65:21–28.
- Kar B, Adkins LE, Civitello AB, et al. Clinical experience with the TandemHeart percutaneous ventricular assist device. *Tex Heart Inst J.* 2006;33:111–115.
- Badu B, Cain MT, Durham 3rd LA, et al. A dual-lumen percutaneous cannula for managing refractory right ventricular failure. *ASAIO J.* 2020;66:915–921.
- Carrozini M, Merlanti B, Olivieri GM, et al. Percutaneous RVAD with the Protek Duo for severe right ventricular primary graft dysfunction after heart transplant. *J Heart Lung Transplant.* 2021;40:580–583.
- Rivera-Lebron B, McDaniel M, Ahrar K, et al. Diagnosis, treatment and follow up of acute pulmonary embolism: consensus practice from the PERT Consortium. *Clin Appl Thromb Hemost.* 2019;25:1076029619853037.
- Cook JA, Shah KB, Quader MA, et al. The total artificial heart. *J Thorac Dis.* 2015;7:2172–2180.
- Goodwin ML, Mokadam NA. Total artificial heart implantation: how i teach it. *Ann Thorac Surg.* 2019;108:1271–1276.
- Stainback RF, Estep JD, Agler DA, et al. Echocardiography in the management of patients with left ventricular assist devices: recommendations from the American Society of Echocardiography. *J Am Soc Echocardiogr.* 2015;28:853–909.
- Charron C, Repesse X, Bodson L, Au SM, Vieillard-Baron A. Ten good reasons why everybody can and should perform cardiac ultrasound in the ICU. *Anaesthesiol Intensive Ther.* 2014;46:319–322.
- Flynn BC, Spellman J, Bodian C, Moitra VK. Inadequate visualization and reporting of ventricular function from transthoracic echocardiography after cardiac surgery. *J Cardiothorac Vasc Anesth.* 2010;24:280–284.
- Schmidt S, Dieks JK, Quintel M, Moerer O. Critical care echocardiography as a routine procedure for the detection and early treatment of cardiac pathologies. *Diagnostics.* 2020;10.
- Tchoukina I, Smallfield MC, Shah KB. Device management and flow optimization on left ventricular assist device support. *Crit Care Clin.* 2018;34:453–463.
- Bouchez S, Van Bellegem Y, De Somer F, De Pauw M, Strooband R, Wouters P. Haemodynamic management of patients with left ventricular assist devices using echocardiography: the essentials. *Eur Heart J Cardiovasc Imaging.* 2019;20:373–382.
- Burzotta F, Trani C, Doshi SN, et al. Impella ventricular support in clinical practice: collaborative viewpoint from a European expert user group. *Int J Cardiol.* 2015;201:684–691.
- Pretorius M, Hughes AK, Stahlman MB, et al. Placement of the TandemHeart percutaneous left ventricular assist device. *Anesth Analg.* 2006;103:1412–1413.
- Mizuguchi KA, Padera Jr RF, Kowalczyk A, Doran MN, Couper GS, Fox AA. Transesophageal echocardiography imaging of the total artificial heart. *Anesth Analg.* 2013;117:780–784.
- Vahdatpour C, Collins D, Goldberg S. Cardiogenic shock. *J Am Heart Assoc.* 2019;8:e011991.
- Smishney NJ, Shaw AD, Stapelfeldt WH, et al. Postoperative hypotension in patients discharged to the intensive care unit after non-cardiac surgery is associated with adverse clinical outcomes. *Crit Care.* 2020;24:682.
- Maheshwari K, Nathanson BH, Munson SH, et al. The relationship between ICU hypotension and in-hospital mortality and morbidity in septic patients. *Intensive Care Med.* 2018;44:857–867.
- Gregory A, Stapelfeldt WH, Khanna AK, et al. Intraoperative hypotension is associated with adverse clinical outcomes after noncardiac surgery. *Anesth Analg.* 2021;132(6):1654–1665.
- Moazami N, Fukumachi K, Kobayashi M, et al. Axial and centrifugal continuous-flow rotary pumps: a translation from pump mechanics to clinical practice. *J Heart Lung Transplant.* 2013;32:1–11.
- Bootsma IT, de Lange F, Scheeren TWL, Jainandunsing JS, Boerma EC. High versus normal blood pressure targets in relation to right ventricular dysfunction after cardiac surgery: a randomized controlled trial. *J Cardiothorac Vasc Anesth.* 2021;35:2980–2990.

41. Tanaka D, Shimada S, Mullin M, Kreitler K, Cavarocchi N, Hirose H. What is the optimal blood pressure on veno-arterial extracorporeal membrane oxygenation? Impact of mean arterial pressure on survival. *ASAIO J*. 2019;65:336–341.
42. Cevasco M, Takayama H, Ando M, Garan AR, Naka Y, Takeda K. Left ventricular distension and venting strategies for patients on venoarterial extracorporeal membrane oxygenation. *J Thorac Dis*. 2019;11:1676–1683.
43. Donker DW, Brodie D, Henriques JPS, Broome M. Left ventricular unloading during veno-arterial ECMO: a review of percutaneous and surgical unloading interventions. *Perfusion*. 2019;34:98–105.
44. Pucher PH, Cummings IG, Shipolini AR, McCormack DJ. Is heparin needed for patients with an intra-aortic balloon pump? *Interact Cardiovasc Thorac Surg*. 2012;15:136–139.
45. Jiang CY, Zhao LL, Wang JA, Mohammad B. Anticoagulation therapy in intra-aortic balloon counterpulsation: does IABP really need anti-coagulation? *J Zhejiang Univ - Sci*. 2003;4:607–611.
46. Reed BN, DiDomenico RJ, Allender JE, et al. Survey of anticoagulation practices with the Impella percutaneous ventricular assist device at high-volume centers. *J Interv Cardiol*. 2019;2019:3791307.
47. Bowers RD, Nowak J, Bryan D, Biondi N, Dorsett B. Utilization of a dextrose-only purge solution for an LVAD in patients with suspected HIT: a case series. *J Pharm Pract*. 2021;34(6):966–969.
48. Szymanski TW, Weeks PA, Lee Y, et al. Anticoagulation of Impella with a bivalirudin purge solution. *ASAIO J*. 2020;66:e117–e120.
49. Fabrizio C, Levito MN, Rivoecchi R, et al. Outcomes of systemic anticoagulation with bivalirudin for Impella 5.0. *Int J Artif Organs*. 2021;44(10):681–686.
50. Lee Y, Weeks PA. Effectiveness of protocol guided heparin anticoagulation in patients with the TandemHeart percutaneous ventricular assist device. *ASAIO J*. 2015;61:207–208.
51. Copeland J, Copeland H, Nolan P, Gustafson M, Slepian M, Smith R. Results with an anticoagulation protocol in 99 SynCardia total artificial heart recipients. *ASAIO J*. 2013;59:216–220.
52. Mohite PN, Zych B, Popov AF, et al. CentriMag short-term ventricular assist as a bridge to solution in patients with advanced heart failure: use beyond 30 days. *Eur J Cardio Thorac Surg*. 2013;44:e310–e315.
53. John R, Long JW, Massey HT, et al. Outcomes of a multicenter trial of the Levitronix CentriMag ventricular assist system for short-term circulatory support. *J Thorac Cardiovasc Surg*. 2011;141:932–939.
54. Wood KL, Ayers B, Gosev I, et al. Venoarterial-extracorporeal membrane oxygenation without routine systemic anticoagulation decreases adverse events. *Ann Thorac Surg*. 2020;109:1458–1466.
55. Loyaga-Rendon RY, Kazui T, Acharya D. Antiplatelet and anticoagulation strategies for left ventricular assist devices. *Ann Transl Med*. 2021;9:521.
56. Feldman D, Pamboukian SV, Teuteberg JJ, et al. The 2013 International Society for Heart and Lung Transplantation guidelines for mechanical circulatory support: executive summary. *J Heart Lung Transplant*. 2013;32:157–187.
57. Renou A, Neuschwander A, Kimmoun A, et al. Modified 4T score for heparin-induced thrombocytopenia diagnosis in VA-ECMO patients. *Intensive Care Med*. 2020;46:1481–1483.
58. Warkentin TE, Sheppard JL, Moore JC, Sigouin CS, Kelton JG. Quantitative interpretation of optical density measurements using PF4-dependent enzyme-immunoassays. *J Thromb Haemost*. 2008;6:1304–1312.
59. Koster A, Faraoni D, Levy JH. Argatroban and bivalirudin for perioperative anticoagulation in cardiac surgery. *Anesthesiology*. 2018;128:390–400.
60. Kiser TH, Fish DN. Evaluation of bivalirudin treatment for heparin-induced thrombocytopenia in critically ill patients with hepatic and/or renal dysfunction. *Pharmacotherapy*. 2006;26:452–460.
61. Schmidt M, Brechot N, Hariri S, et al. Nosocomial infections in adult cardiogenic shock patients supported by venoarterial extracorporeal membrane oxygenation. *Clin Infect Dis*. 2012;55:1633–1641.
62. Frerou A, Lesouhaitier M, Gregoire M, et al. Venoarterial extracorporeal membrane oxygenation induces early immune alterations. *Crit Care*. 2021;25:9.
63. Menaker J, Galvagno S, Rabinowitz R, et al. Epidemiology of blood stream infection in adult extracorporeal membrane oxygenation patients: a cohort study. *Heart Lung*. 2019;48:236–239.
64. Aslam S, Xie R, Cowger J, et al. Bloodstream infections in mechanical circulatory support device recipients in the international society of heart and lung transplantation mechanically assisted circulation support registry: epidemiology, risk factors, and mortality. *J Heart Lung Transplant*. 2018;37:1013–1020.
65. Tempelhof MW, Klein L, Cotts WG, et al. Clinical experience and patient outcomes associated with the TandemHeart percutaneous transeptal assist device among a heterogeneous patient population. *ASAIO J*. 2011;57:254–261.
66. O'Horo JC, Cawcutt KA, De Moraes AG, Sampathkumar P, Schears GJ. The evidence base for prophylactic antibiotics in patients receiving extracorporeal membrane oxygenation. *ASAIO J*. 2016;62:6–10.
67. Li M, Mazzeffi MA, Gammie JS, et al. Characterization of postoperative infection risk in cardiac surgery patients with delayed sternal closure. *J Cardiothorac Vasc Anesth*. 2020;34:1238–1243.
68. de Jonge SW, Boldingh QJ, Solomkin JS, et al. Effect of postoperative continuation of antibiotic prophylaxis on the incidence of surgical site infection: a systematic review and meta-analysis. *Lancet Infect Dis*. 2020;20:1182–1192.
69. Branch-Elliman W, O'Brien W, Strymish J, Itani K, Gupta K. Association of duration and type of surgical prophylaxis with antimicrobial-associated adverse events. *JAMA Surg*. 2019;154:590–598.
70. Plewes K, Kingston HW, Ghose A, et al. Dondorp AM: cell-free hemoglobin mediated oxidative stress is associated with acute kidney injury and renal replacement therapy in severe falciparum malaria: an observational study. *BMC Infect Dis*. 2017;17:313.
71. Vermeulen Windsant IC, Snoeijs MG, Hanssen SJ, et al. Hemolysis is associated with acute kidney injury during major aortic surgery. *Kidney Int*. 2010;77:913–920.
72. Reiter CD, Wang X, Tanus-Santos JE, et al. Cell-free hemoglobin limits nitric oxide bioavailability in sickle-cell disease. *Nat Med*. 2002;8:1383–1389.
73. Lyu L, Long C, Hei F, et al. Plasma free hemoglobin is a predictor of acute renal failure during adult venous-arterial extracorporeal membrane oxygenation support. *J Cardiothorac Vasc Anesth*. 2016;30:891–895.
74. Omar HR, Mirsaedi M, Socias S, et al. Plasma free hemoglobin is an independent predictor of mortality among patients on extracorporeal membrane oxygenation support. *PLoS One*. 2015;10:e0124034.
75. Esposito ML, Morine KJ, Annamalai SK, et al. Increased plasma-free hemoglobin levels identify hemolysis in patients with cardiogenic shock and a trans valvular micro-axial flow pump. *Artif Organs*. 2019;43:125–131.
76. Permerfeldt CE, Toma A, Badheka AO, et al. Severe hemolytic anemia and acute kidney injury after percutaneous continuous-flow ventricular assistance. *Circ Heart Fail*. 2011;4:e20–e22.
77. Badiye AP, Hernandez GA, Novoa I, Chaparro SV. Incidence of hemolysis in patients with cardiogenic shock treated with impella percutaneous left ventricular assist device. *ASAIO J*. 2016;62:11–14.
78. Chan CH, Pieper IL, Hambly R, et al. The CentriMag centrifugal blood pump as a benchmark for in vitro testing of hemocompatibility in implantable ventricular assist devices. *Artif Organs*. 2015;39:93–101.
79. Roka-Moia Y, Li M, Ivich A, Muslmani S, Kern KB, Slepian MJ. Impella 5.5 versus Centrimag: a head-to-head comparison of device hemocompatibility. *ASAIO J*. 2020;66:1142–1151.
80. Shin DA, Lee JC, Shin H, Cho YJ, Kim HC. Point-of-care testing of plasma free hemoglobin and hematocrit for mechanical circulatory support. *Sci Rep*. 2021;11:3788.
81. Rimsans J, Sylvester KW, Connors JM. Direct Thrombin inhibitor for LVAD thrombosis: a closer look. *Clin Appl Thromb Hemost*. 2017;23:405–409.
82. Dang G, Epperla N, Muppidi V, et al. Medical management of pump-related thrombosis in patients with continuous-flow left ventricular assist devices: a systematic review and meta-analysis. *ASAIO J*. 2017;63:373–385.
83. Wetz AJ, Brauer A, Quintel M, Heise D. Does sodium bicarbonate infusion really have no effect on the incidence of acute kidney injury after cardiac surgery? A prospective observational trial. *Crit Care*. 2015;19:183.
84. Haase M, Haase-Fielitz A, Bellomo R. Cardiopulmonary bypass, hemolysis, free iron, acute kidney injury and the impact of bicarbonate. *Contrib Nephrol*. 2010;165:28–32.
85. Mazzeffi M, Greenwood J, Tanaka K, et al. Bleeding, transfusion, and mortality on extracorporeal life support: ECLS Working Group on Thrombosis and Hemostasis. *Ann Thorac Surg*. 2016;101:682–689.
86. Mincu RI, Rassaf T, Totzeck M. Red blood cell transfusion in patients with ST-elevation myocardial infarction: a meta-analysis of more than 21,000 patients. *Neth Heart J*. 2018;26:454–460.
87. Zhang W, Zheng Y, Yu K, Gu J. Liberal transfusion versus restrictive transfusion and outcomes in critically ill adults: a meta-analysis. *Transfus Med Hemother*. 2021;48:60–68.
88. Kalbhenn J, Schlagenhauf A, Rosenfelder S, Schmutz A, Zieger B. Acquired von Willebrand syndrome and impaired platelet function during venovenous extracorporeal membrane oxygenation: rapid onset and fast recovery. *J Heart Lung Transplant*. 2018;37:985–991.
89. Tauber H, Ott H, Streif W, et al. Extracorporeal membrane oxygenation induces short-term loss of high-molecular-weight von Willebrand factor multimers. *Anesth Analg*. 2015;120:730–736.
90. Chen Z, Zhang J, Li T, Tran D, Griffith BP, Wu ZJ. The impact of shear stress on device-induced platelet hemostatic dysfunction relevant to thrombosis and bleeding in mechanically assisted circulation. *Artif Organs*. 2020;44:E201–E213.
91. Chen Z, Mondal NK, Ding J, Gao J, Griffith BP, Wu ZJ. Shear-induced platelet receptor shedding by non-physiological high shear stress with short exposure time: glycoprotein Ib/alpha and glycoprotein VI. *Thromb Res*. 2015;135:692–698.
92. Chen Z, Koening SC, Slaughter MS, Griffith BP, Wu ZJ. Quantitative characterization of shear-induced platelet receptor shedding: glycoprotein Iba, glycoprotein VI, and glycoprotein IIb/IIIa. *ASAIO J*. 2018;64:773–778.
93. Raphael J, Mazer CD, Wilkey A, et al. Corrigendum to 'society of cardiovascular anesthesiologists (SCA) clinical practice improvement (CPI) advisory for management of perioperative bleeding and hemostasis in cardiac surgery patients'. *J Cardiothorac Vasc Anesth*. 2019;33:2887–2899. *J Cardiothorac Vasc Anesth*. 2020; 34: 840–841.
94. Tibi P, McClure RS, Huang J, et al. STS/SCA/AmSECT/SABM update to the clinical practice guidelines on patient blood management. *J Cardiothorac Vasc Anesth*. 2021;35:2569–2591.
95. Mazzeffi M, Hasan S, Abuelkasem E, et al. Von Willebrand factor-GP1b interactions in venoarterial extracorporeal membrane oxygenation patients. *J Cardiothorac Vasc Anesth*. 2019;33:2125–2132.
96. Bui JD, Despotis GD, Trulock EP, Patterson GA, Goodnough LT. Fatal thrombosis after administration of activated prothrombin complex concentrates in a patient supported by extracorporeal membrane oxygenation who had received activated recombinant factor VII. *J Thorac Cardiovasc Surg*. 2002;124:852–854.
97. Repesse X, Au SM, Brechot N, et al. Recombinant factor VIIa for uncontrollable bleeding in patients with extracorporeal membrane oxygenation: report on 15 cases and literature review. *Crit Care*. 2013;17:R55.
98. Brown CS, Zemrak WR, Dyer KJ, Rolfe S. Low-dose prothrombin complex concentrate in patients with left ventricular assist devices. *ASAIO J*. 2019;65:e21–e24.
99. Wong CW. Anticoagulation for stroke prevention in elderly patients with non-valvular atrial fibrillation: what are the obstacles? *Hong Kong Med J*. 2016;22:608–615.
100. Kadado AJ, Akar JG, Hummel JP. Arrhythmias after left ventricular assist device implantation: incidence and management. *Trends Cardiovasc Med*. 2018;28:41–50.
101. Ranganath NK, Smith DE, Moazami N. The Achilles' heel of left ventricular assist device therapy: right ventricle. *Curr Opin Organ Transplant*. 2018;23:295–300.
102. Long B, Robertson J, Koefman A, Brady W. Left ventricular assist devices and their complications: a review for emergency clinicians. *Am J Emerg Med*. 2019;37:1562–1570.
103. Snipelsky D, Reddy YN, Manocha K, et al. Effect of ventricular arrhythmia ablation in patients with Heart Mate II left ventricular assist devices: an evaluation of ablation therapy. *J Cardiovasc Electro-physiol*. 2017;28:68–77.
104. Ahmed A, Amin M, Boilson BA, Killu AM, Madhavan M. Ventricular arrhythmias in patients with left ventricular assist device (LVAD). *Curr Treat Options Cardiovasc Med*. 2019;21:75.
105. Williams B, Bernstein W. Review of venoarterial extracorporeal membrane oxygenation and development of intracardiac thrombosis in adult cardiothoracic patients. *J Extra Corpor Technol*. 2016;48:162–167.
106. Palaniswamy C, Miller MA, Reddy VY, Dukkupati SR. Hemodynamic support for ventricular tachycardia ablation. *Card Electrophysiol Clin*. 2017;9:141–152.
107. Wehman B, Mazzeffi M, Chow R, et al. Thoracoscopic sympathectomy for refractory electrical storm after coronary artery bypass grafting. *Ann Thorac Surg*. 2018;105:e99–e101.
108. Chatterjee NA, Upadhyay GA, Ellenbogen KA, McAlister FA, Choudhry NK, Singh JP. Atrioventricular nodal ablation in atrial fibrillation: a meta-analysis and systematic review. *Circ Arrhythm Electrophysiol*. 2012;5:68–76.
109. Aggarwal S, Bannan S. Displacement of impella post chest compressions. *Heart Views*. 2014;15:127–128.
110. Haji JY, Mehra S, Doraiswamy P. Awake ECMO and mobilizing patients on ECMO. *Indian J Thorac Cardiovasc Surg*. 2021;37(suppl 2):309–318.
111. Parry SM, Puthucherry ZA. The impact of extended bed rest on the musculoskeletal system in the critical care environment. *Extrem Physiol Med*. 2015;4:16.
112. Hermans G, Van den Bergh G. Clinical review: intensive care unit acquired weakness. *Crit Care*. 2015;19:274.
113. Engel HJ, Needham DM, Morris PE, Gropper MA. ICU early mobilization: from recommendation to implementation at three medical centers. *Crit Care Med*. 2013;41:S69–S80.
114. Engel HJ, Tatebe S, Alonzo PB, Mustille RL, Rivera MJ. Physical therapist-established intensive care unit early mobilization program: quality improvement project for critical care at the University of California San Francisco Medical Center. *Phys Ther*. 2013;93:975–985.
115. Elouze O, Lamirel J, Perrot J, et al. Extubation of patients undergoing extracorporeal life support. A retrospective study. *Perfusion*. 2019;34:50–57.
116. Pasirja C, Mackowick KM, Rathel M, et al. Ambulation with femoral arterial cannulation can be safely performed on venoarterial extracorporeal membrane oxygenation. *Ann Thorac Surg*. 2019;107:1389–1394.
117. Wells CL, Forrester J, Vogel J, Rector R, Tabatabai A, Herr D. Safety and feasibility of early physical therapy for patients on extracorporeal membrane oxygenator: university of Maryland Medical Center experience. *Crit Care Med*. 2018;46:53–59.

118. Ferreira DDC, Marcolino MAZ, Macagnan FE, Plentz RDM, Kessler A. Safety and potential benefits of physical therapy in adult patients on extracorporeal membrane oxygenation support: a systematic review. *Rev Bras Ter Intensiva*. 2019;31:227–239.
119. Park J, Heo E, Song IA, et al. Nutritional support and clinical outcomes in critically ill patients supported with veno-arterial extracorporeal membrane oxygenation. *Clin Nutr*. 2020;39:2617–2623.
120. McClave SA, Taylor BE, Martindale RG, Society of Critical Care M, American Society for P, Enteral N, et al. Guidelines for the provision and assessment of nutrition support therapy in the adult critically ill patient: society of Critical Care Medicine (SCCM) and American Society for Parenteral and Enteral Nutrition (A.S.P.E.N.). *JPEN J Parenter Enteral Nutr*. 2016;40:159–211.
121. Davies AR, Morrison SS, Bailey MJ, et al. A multicenter, randomized controlled trial comparing early nasojejunal with nasogastric nutrition in critical illness. *Crit Care Med*. 2012;40:2342–2348.
122. Mehra MR, Canter CE, Hannan MM, et al. The 2016 International Society for Heart Lung Transplantation listing criteria for heart transplantation: a 10-year update. *J Heart Lung Transplant*. 2016;35:1–23.
123. Yancy CW, Jessup M, Bozkurt B, et al. 2017 ACC/AHA/HFSA focused update of the 2013 ACCF/AHA guideline for the management of heart failure: a report of the American college of cardiology/American heart association task force on clinical practice guidelines and the heart failure society of America. *J Am Coll Cardiol*. 2017;70:776–803.
124. Choi KH, Choi JO, Jeon ES, et al. Guideline-directed medical therapy for patients with heart failure with midrange ejection fraction: a patient-pooled analysis from the Kor HF and Kor AHF Registries. *J Am Heart Assoc*. 2018;7:e009806.
125. Joseph J, P SS, James J, Abraham S, Abdullakutty J. Guideline-directed medical therapy in heart failure patients: impact of focused care provided by a heart failure clinic in comparison to general cardiology out-patient department. *Egypt Heart J*. 2020;72:53.
126. Lichtenthal WG, Kissane DW. The management of family conflict in palliative care. *Prog Palliat Care*. 2008;16:39–45.